EMBEDDED SYSTEM

WHAT IS EMBEDDED SYSTEM?

Embedded system = Hardware + Software

Designed for specific purpose

An embedded system is a combination of computer hardware and software, designed for a specific function or functions within a larger system.

WHAT IS EMBEDDED SYSTEM?

S/W

```
#include "mbed.h"
#include "C12832.h"
#include "Sht31.h"

C12832 lcd(SPI_MOSI, SPI_SCK, SPI_MISO, p8, p11);
Sht31 sht31(I2C_SDA, I2C_SCL);
DigitalOut led(LED1);

int main() {
    printf("Set the temperature above 25 degrees to trigger the warning LED\n");

    while (1) {
        lcd.cls();

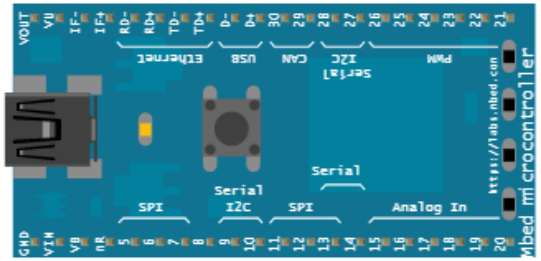
        float temp = sht31.readTemperature();
        float humidity = sht31.readHumidity();

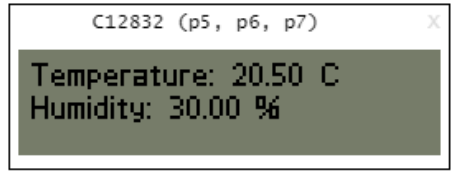
        lcd.locate(3, 3);
        lcd.printf("Temperature: %.2f C", temp);
        lcd.locate(3, 13);
        lcd.printf("Humidity: %.2f %%", humidity);

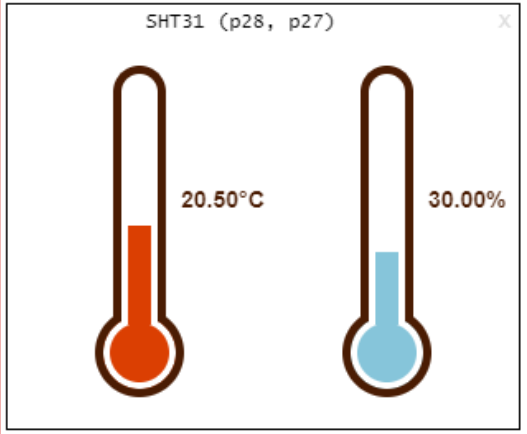
        // turn on LED if the temperature is above 25 degrees
        led = temp > 25.0f;

        wait(0.5f);
    }
}
```

H/W







Embedded System

WHAT IS EMBEDDED SYSTEM?



Industrial Robots



GPS Receivers



Digital Cameras



DVD Players

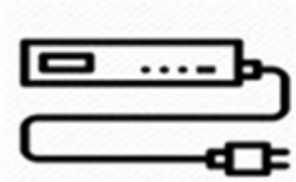


Wireless Routers

Embeeded System = *Computer Inside a Product*



MP3 Players



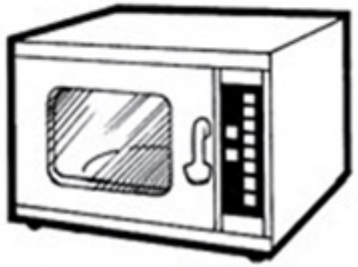
Set top Boxes



Gaming Consoles

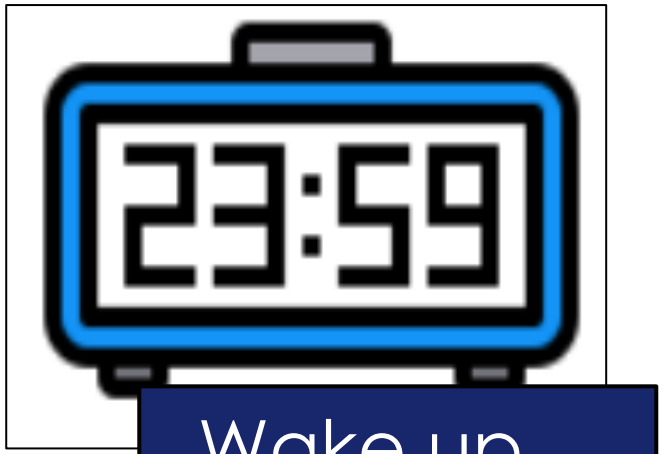


Photocopiers



Microwave Ovens

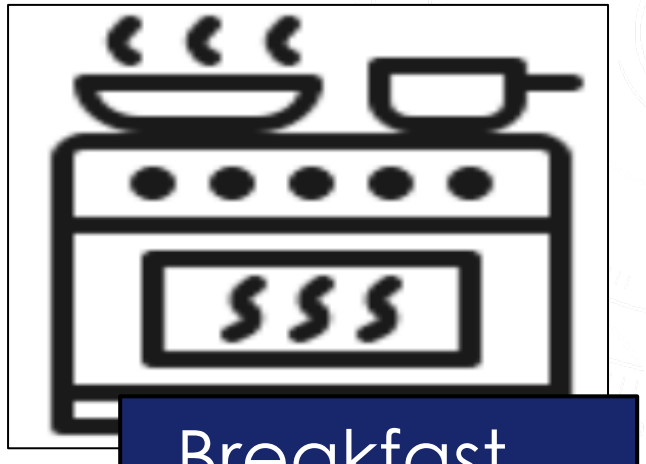
Embedded system in daily life....



Wake up...



Washing...



Breakfast...



Elevator..



Car...

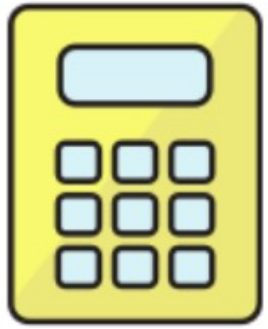


Traffic lights...

Inside a classroom…



Smart watch…



Calculator…



Projector…



Mobile phone…



Smart shoe…

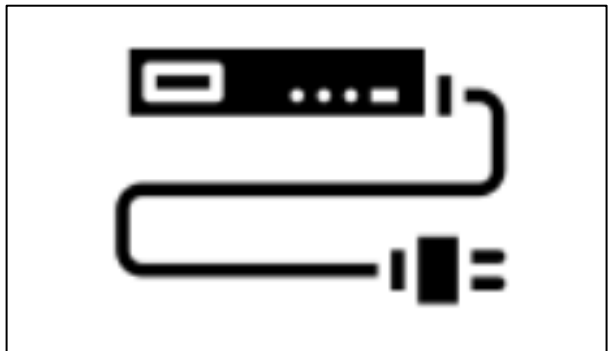


Wireless mic…

Back home.....



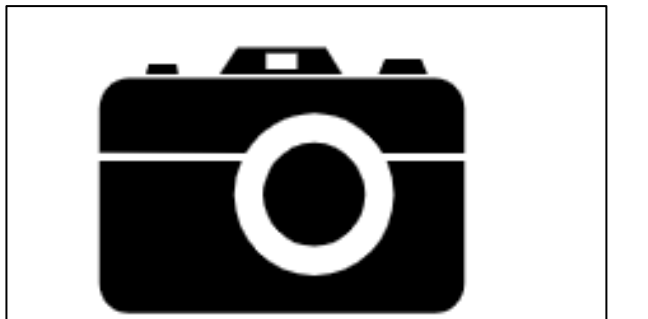
Smart TV…



Set top box…



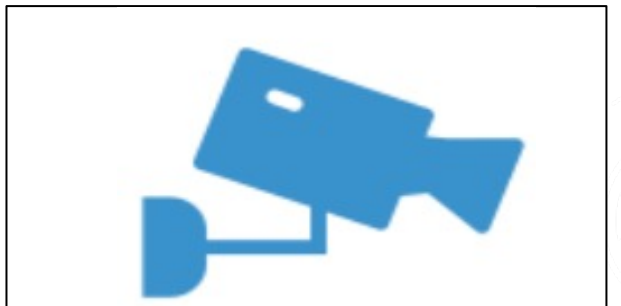
Headphones…



Digital Camera…

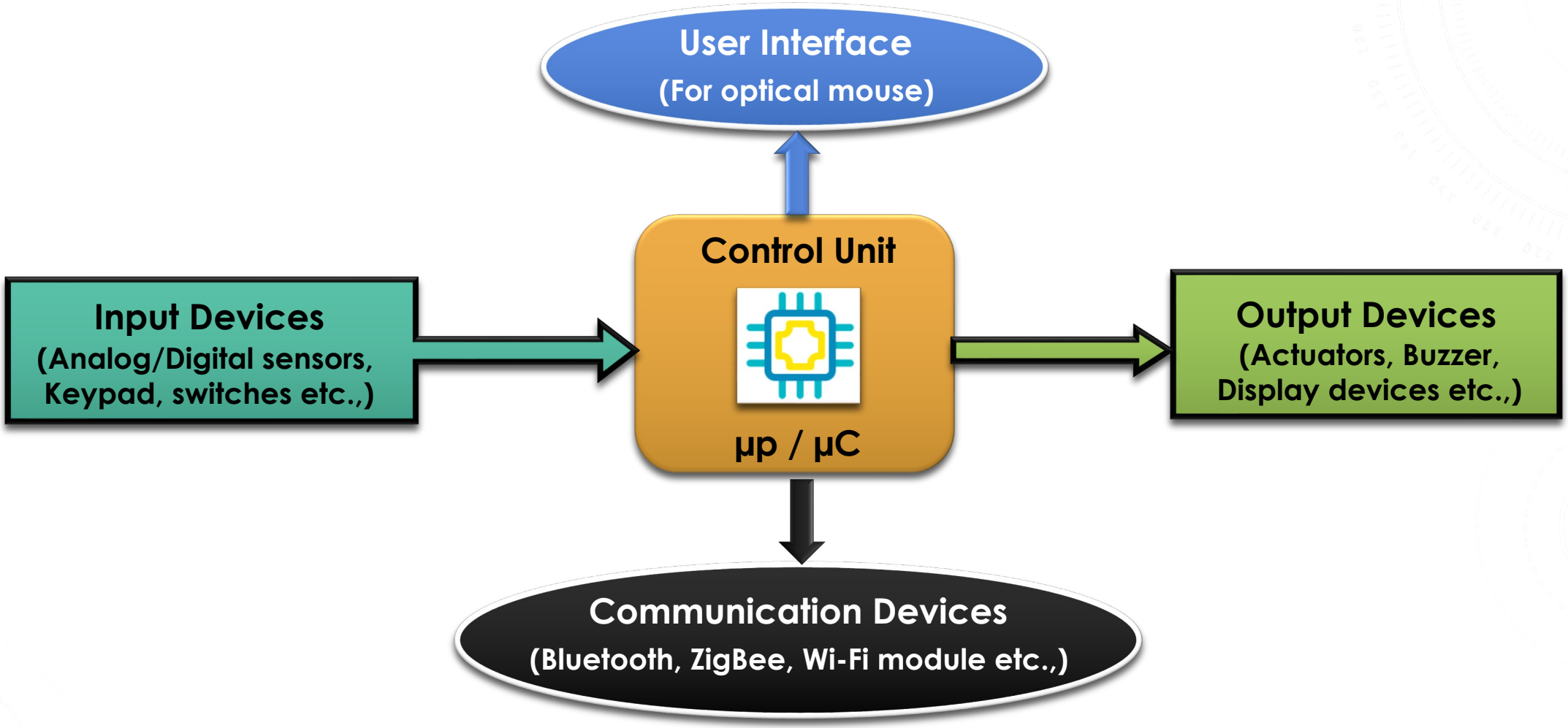


Printer…



CCTV…

EMBEDDED SYSTEM – BLOCK DIAGRAM



EMBEDDED SYSTEM - EXAMPLES

PURPOSE OF AN EMBEDDED SYSTEM

- Each embedded system is designed to serve the purpose of any one or combinations of following task:
  - Data collection / storage / representation
  - Data communication
  - Data (signal) processing
  - Monitoring
  - Controlling
  - Application specific user interface

EMBEDDED SYSTEM - EXAMPLES

MAJOR APPLICATION DOMAINS OF EMBEDDED SYSTEM

- Home Appliances
- Consumer Electronics
- Home Automation / Security
- Computer components
- Automotive
- Health care
- Military
- Industrial
- Business
- Retail
- Communications
- Avionics
- Banking

EMBEDDED SYSTEM - CHARACTERISTICS

EMBEDDED SYSTEM - CHARACTERISTICS

- Dedicated to specific tasks.
- Supported by a wide array of processors and processor architectures.
- Usually cost sensitive.
- Real-time constraints.
- Require specialized tools and methods.

EMBEDDED SYSTEM - CHARACTERISTICS

- Software failure is much more severe.
- Power & memory constraints.
- Operate under extreme environmental conditions.
- Fewer system resources.
- Real Time Operating System

WHY IS EMBEDDED SYSTEM DIFFERENT?

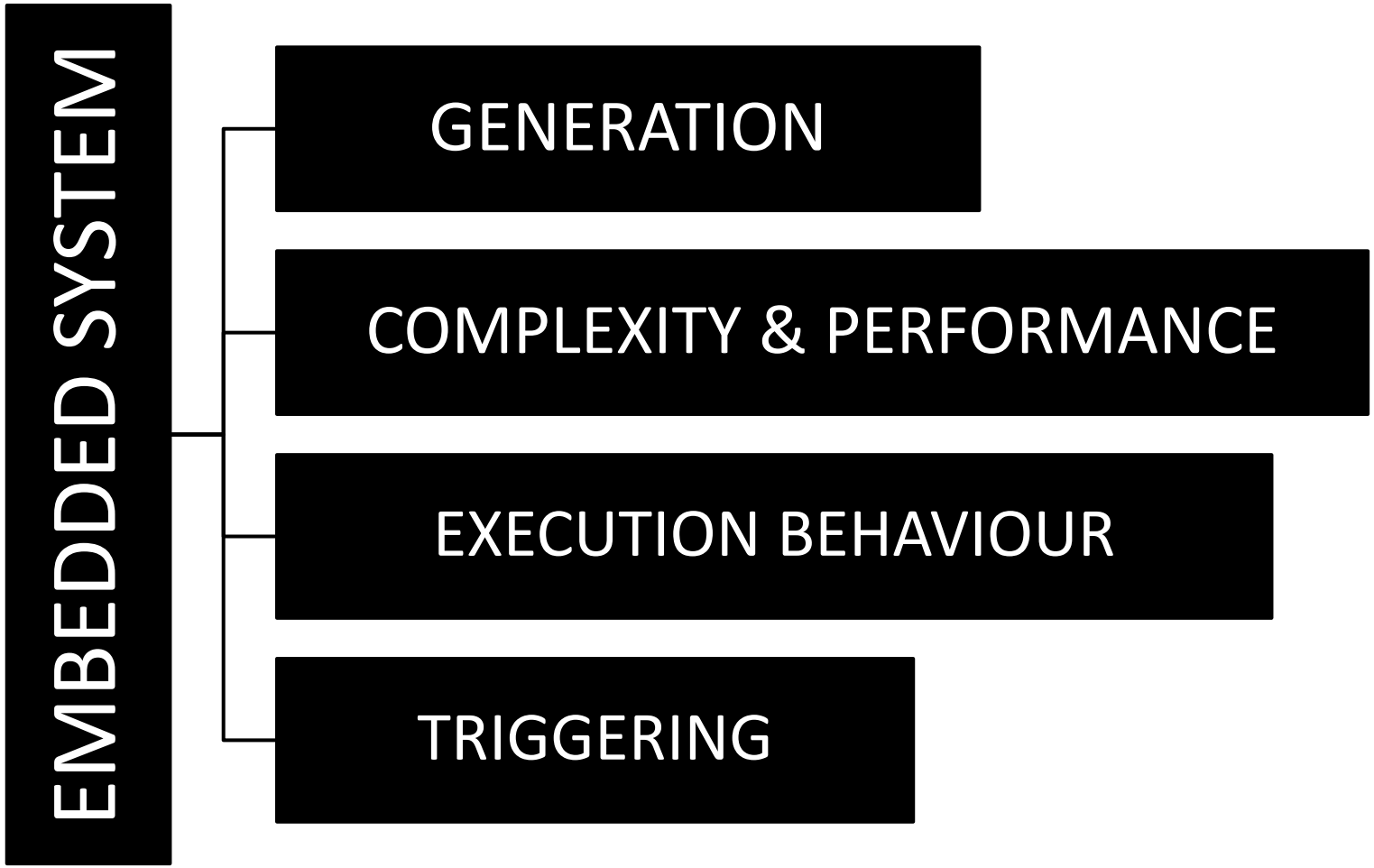
- Complex algorithms – (Automotive Engine ECU)
- User interface – (GPS)
- Real time – (ABS)
- Multirate – (Multimedia Application)
- Manufacturing cost – (Mobile Phone)
- Power and energy – (Data Logger)

GENERAL PURPOSE vs EMBEDDED SYSTEM

S. NO.	GENERAL PURPOSE COMPUTING SYSTEM	EMBEDDED SYSTEM
1	A system which is a combination of generic hardware and general purpose operating system for executing verity of applications	A system which is a combination of special purpose hardware and embedded OS for executing a specific set of application
2	Contains General Purpose Operating System (GPOS)	May or may not contain operating system for function
3	Applications can be modified by user	The firmware of the embedded system is pre-programmed and it is non-modifiable by the end user
4	Performance is a key deciding factor in the selection of the system. Always “Faster is better”.	Application specific requirements (like performance, power, memory, cost) are key deciding factor
5	Need not to be deterministic in execution behavior	Execution behavior is deterministic for certain type of system like “hard real time systems” (Missile guidance or airbag system)
6	Personal computer, Laptops	DVD player, Digital camera

EMBEDDED SYSTEM - CLASSIFICATIONS

EMBEDDED SYSTEM - CLASSIFICATIONS



EMBEDDED SYSTEM - CLASSIFICATIONS

BASED ON GENERATION

- First Generation:
  - build around 4-bit and 8-bit processor like 8085, Z80
  - Simple hardware & firmware developed in assembly code
  - Ex: Digital telephone keypads, calculator etc.,
- Second Generation:
  - build around 8-bit and 16-bit processor
  - complex and powerful instruction set
  - Ex: Data acquisition system, SCADA system etc.,

EMBEDDED SYSTEM - CLASSIFICATIONS

BASED ON GENERATION

- **Third Generation:**
  - build around 16-bit and 32-bit processor
  - application specific processor DSP & ASIC were introduced
  - complex and powerful instruction set with pipelining
  - dedicated embedded real-time OS entered into market
  - Ex: robotics, industrial process control, networking etc.,

EMBEDDED SYSTEM - CLASSIFICATIONS

BASED ON GENERATION

- **Fourth Generation:**
  - advent of SOC, reconfigurable & multicore processors
  - high performance, tight integration and miniaturization μP
  - High performance RTOS for functioning
  - **Ex: smart phone, PDA etc.,**

EMBEDDED SYSTEM - CLASSIFICATIONS

BASED ON COMPLEXITY AND SYSTEM PERFORMANCE

- **Small-scale:**
  - simple in application needs
  - performance requirements are not time critical
  - build around low cost & low performance 8 or 16-bit *µp /µC*
  - **Ex: Electronic toys, digital multimeters etc.,**

EMBEDDED SYSTEM - CLASSIFICATIONS

BASED ON COMPLEXITY AND SYSTEM PERFORMANCE

- **Medium-scale:**
  - slightly complex in hardware & software requirements
  - build on medium performance & low cost 16/32-bit *µp /µC*
  - usually contains embedded operating system
  - **Ex: DVD players, network routers etc.,**

EMBEDDED SYSTEM - CLASSIFICATIONS

BASED ON COMPLEXITY AND SYSTEM PERFORMANCE

- **Large-scale:**
  - highly complex hardware and software requirement
  - employed in mission critical application demanding high performance
  - build around high performance 32 or 64-bit µp /µC or SOC or PLDs or multi-core processors
  - may contain complex high performance RTOS
  - **Ex: Missile navigation system, MRI scanner, ABS etc.,**

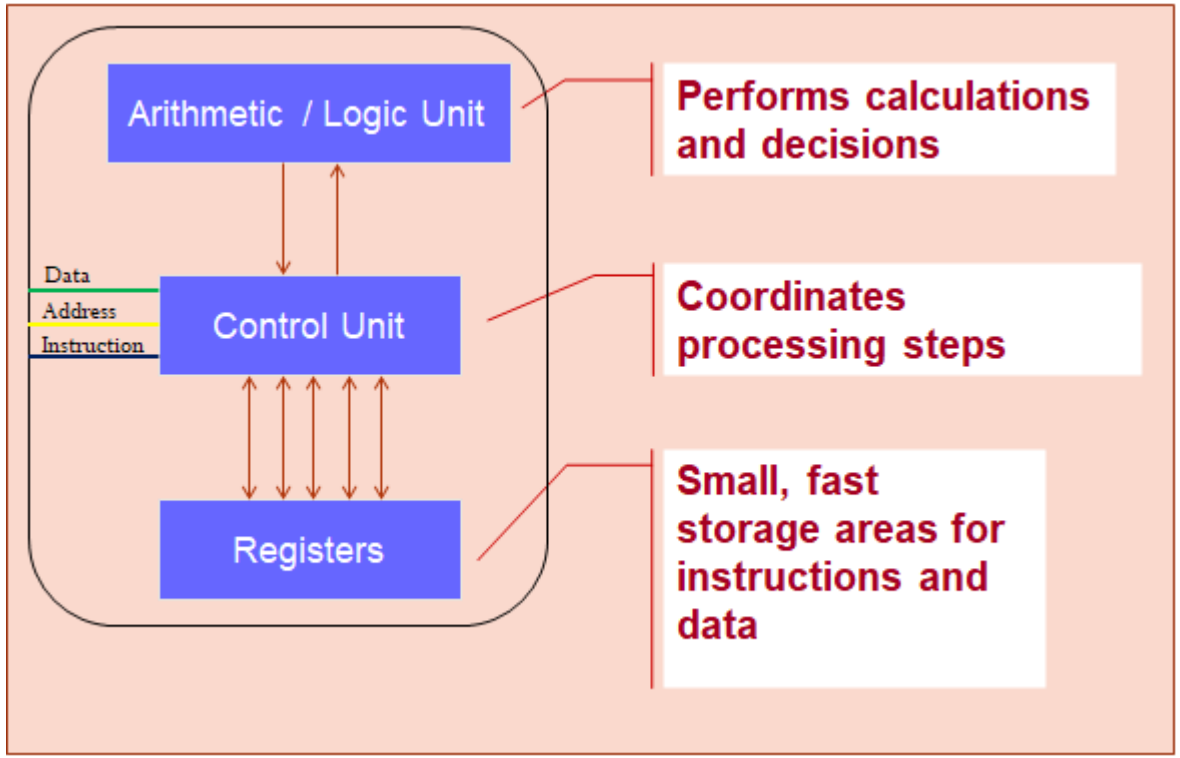
EMBEDDED SYSTEM - CLASSIFICATIONS

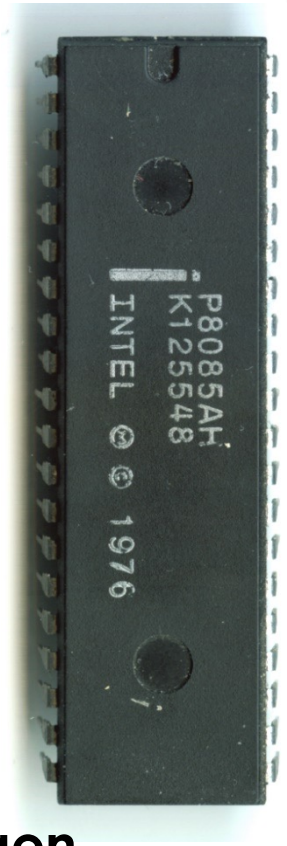
BASED ON EXECUTION BEHAVIOUR

- **Deterministic:**
  - Real-time systems
  - Non Real-time systems
- **Triggering mechanism:**
  - event triggered
  - time triggered

WHAT IS MICROPROCESSOR?

It is a central processing unit (CPU) on a single integrated circuit (IC) chip containing millions of very small components including transistors, resistors, and diodes that work together.





Examples: 8085, 8086, Intel i series, AMD, ARM, Qualcomm Snapdragon

WHAT IS THE FIRST MICROPROCESSOR?

History behind first microprocessor?

Intel 4004



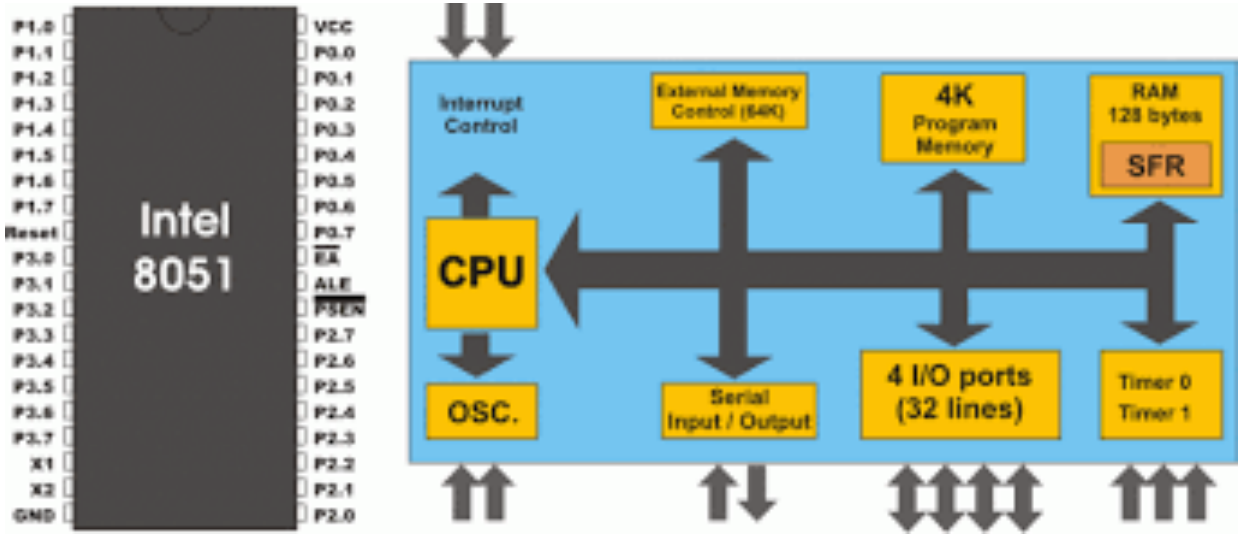
- **1969: The assignment -** *Nippon Calculating Machine Corporation approached Intel to design 12 custom chips for its new Busicom 141-PF* printing calculator.*
- **The Intel solution -** *Intel designed a set of four chips known as the MCS-4. It included* ***CPU(4004),*** *Shift registers(4003) RAM(4002), ROM(4001)*
- **1971: Era of integrated electronics -** *Intel purchased the rights from Nippon Calculating Machine Corporation and launched the Intel 4004 processor and its chipset with an advertisement in the November 15, 1971*

APPLICATIONS OF MICROPROCESSOR

- Laptop
- Servers
- Networking devices
- Desktops
- Mobile phones
- Games machine
- Thin clients
- Military applications

WHAT IS MICROCONTROLLER?

- A microcontroller is a compact integrated circuit designed to govern a specific operation in an embedded system.
- A typical microcontroller includes a processor, memory and input/output (I/O) peripherals on a single chip.





Examples: Intel 8051, PIC 16F877, AVR32, ARM LPC2148

COMMONLY USED MICROCONTROLLER



Atmel AVR



AVR



ATX Mega



ATmega 328P



PIC 18F877A



8051

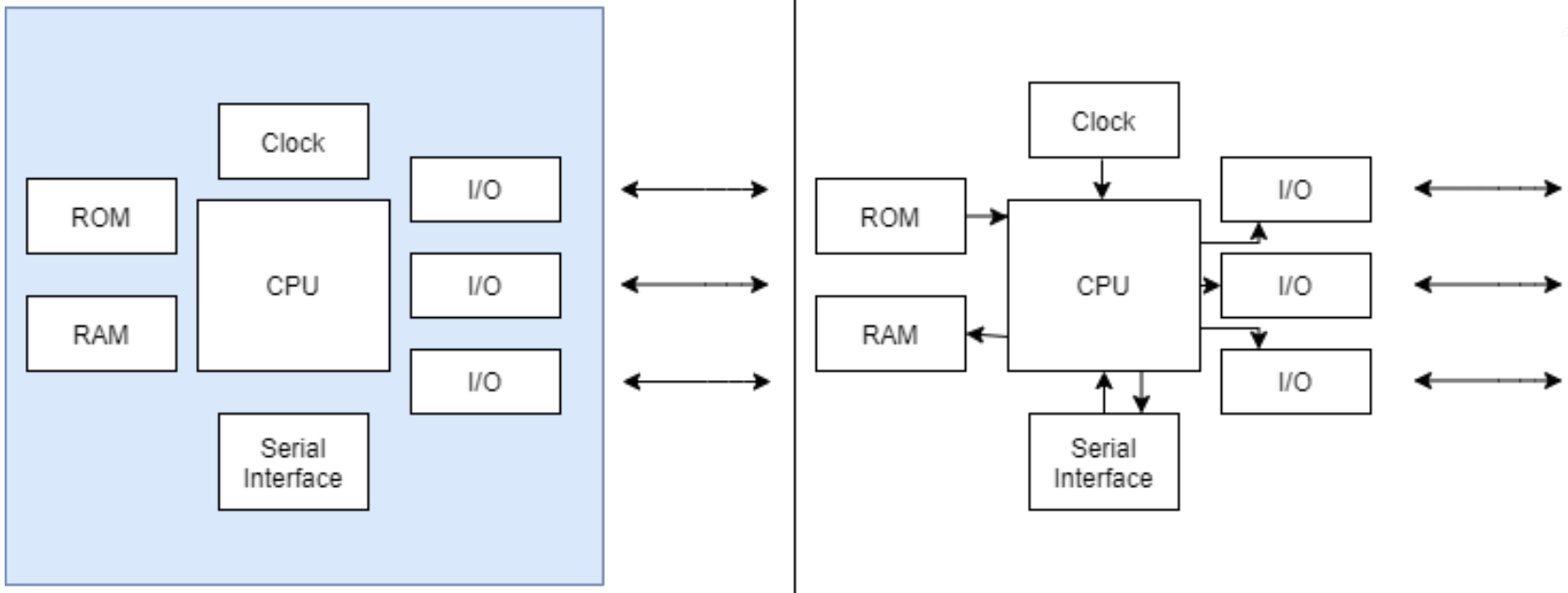


Arduino



ARM

MP vs MC

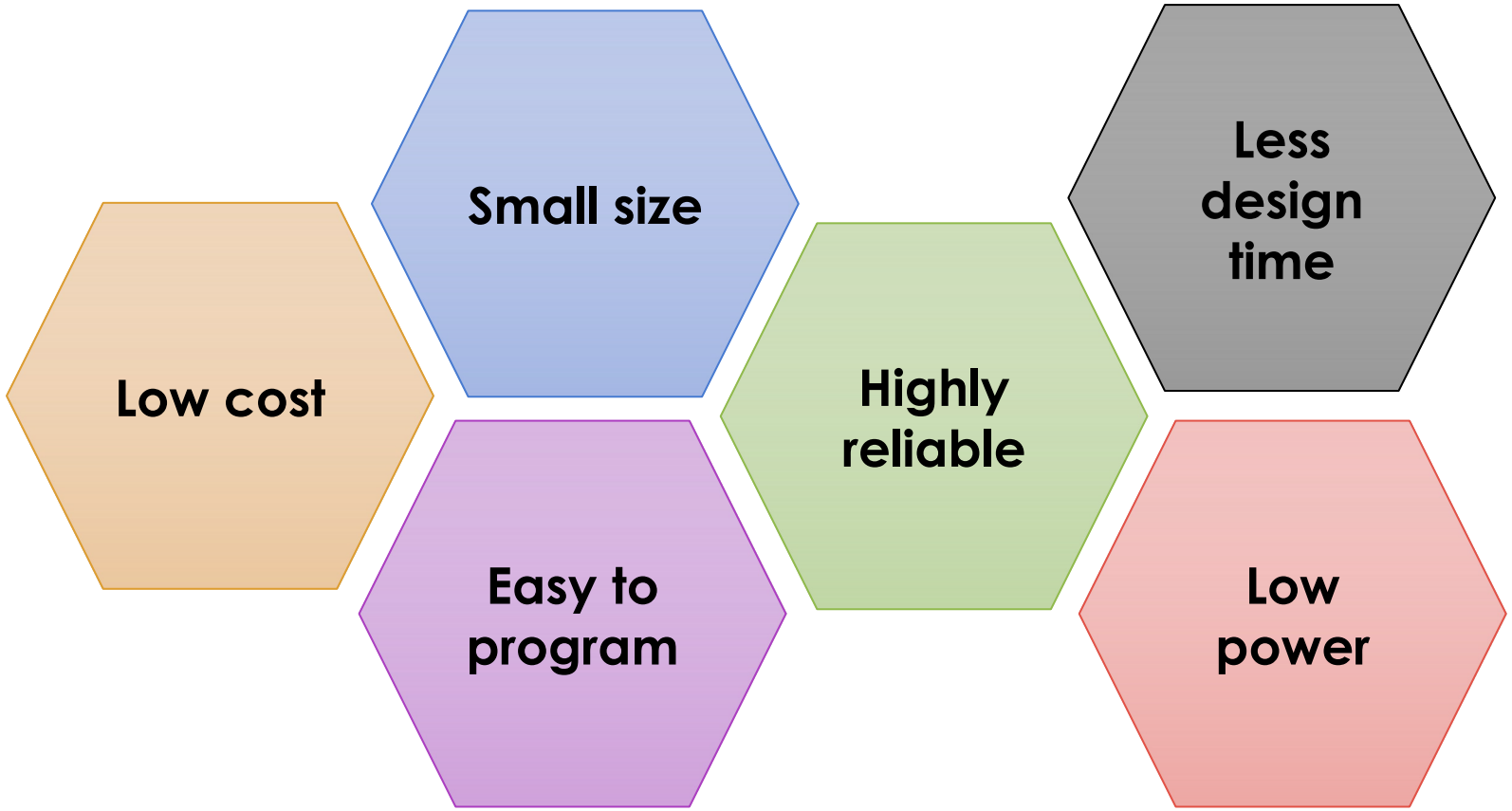


Microcontroller - CPU and other pheripheral in a single chip

Micropocessor - CPU and several supporting chip

WHY MICROCONTROLLERS IN ES?

Most of the embedded system uses Microcontrollers than Microprocessor !!!

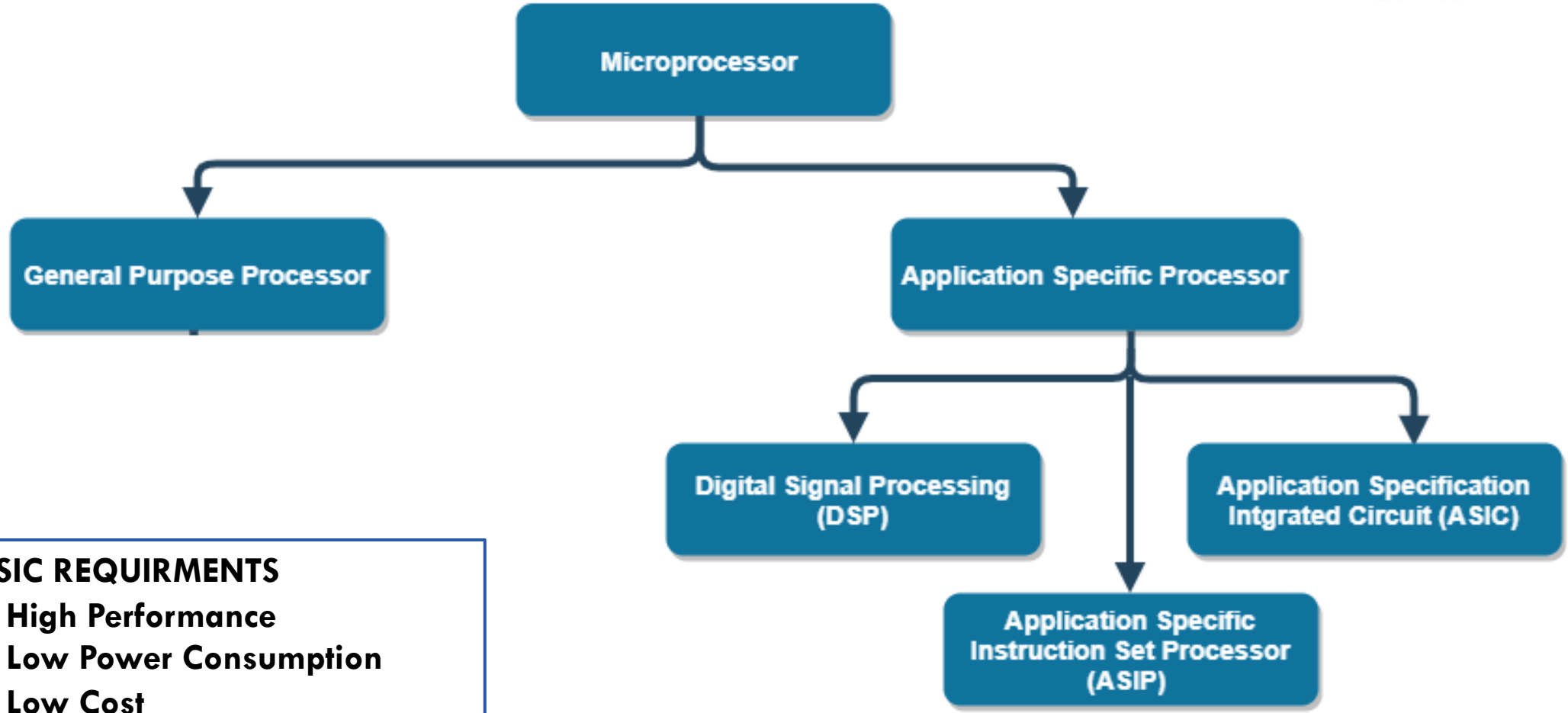


APPLICATIONS OF MICROCONTROLLER

- Automotive
- Medical electronics
- Robotics
- Consumer electronics
- Security (e-commerce, smart cards)
- Industrial automation systems

CLASSIFICATION OF MICROPROCESSORS

CLASSIFICATIONS OF MICROPROCESSORS



BASIC REQUIRMENTS

1. High Performance
2. Low Power Consumption
3. Low Cost

CLASSIFICATIONS OF MICROPROCESSORS

GENERAL PURPOSE PROCESSOR (GPP)

- Programmable device used in a variety of Applications
- GPP are designed to perform multiple tasks
- The system designer only needs to program the processor's memory to carry out the required functionality
- Biggest advantage of such system is the flexibility
- GPP are lack high performance that a certain task required

CLASSIFICATIONS OF MICROPROCESSORS

GENERAL PURPOSE PROCESSOR (GPP)

- Advantages:
  - easy to design and use
  - design time & cost is low
  - reprogrammability
- Disadvantages:
  - performance is not very good
  - size is high
  - they consume much power
- Intel "Pentium" is the most well-known example

CLASSIFICATIONS OF MICROPROCESSORS

APPLICATION SPECIFIC PROCESSORS (ASP)

- ASPs emerged as a solution for high performance and cost effective processors.
- This processor is designed to execute exactly one program
- Contains only the components needed to execute a single program
- An embedded designer creates a single-purpose processor by designing a custom digital circuit

CLASSIFICATIONS OF MICROPROCESSORS

APPLICATION SPECIFIC PROCESSORS (ASP)

- **Advantages:**
  - performance is very good
  - small size (exact to fit one solution)
  - they consume little power
- **Disadvantages:**
  - not very easy to design and design time is thus high
  - design cost is thus higher
  - reprogramming is difficult
  - limited flexibility: not easy to make changes, accommodate features

CLASSIFICATIONS OF MICROPROCESSORS

APPLICATION SPECIFIC PROCESSORS (ASP) - CLASSIFICATIONS

- **Digital Signal Processor (DSP):** Programmable microprocessor for extensive real-time mathematical computations.
- **Application Specific Instruction Set Processors (ASIP):** Programmable microprocessor where hardware and instruction set are designed together for one special application.
- **Application Specific Integrated Circuit (ASIC):** Algorithm completely implemented in hardware.

CLASSIFICATIONS OF MICROPROCESSORS

DIGITAL SIGNAL PROCESSORS (DSP)

- DSPs are specialized microprocessors optimized for the need of performing digital signal processing.
- DSP gained their importance with the increased demand on data intensive applications such as video and internet browsing on mobile devices.
- The digital signal processors satisfy the need for powerful processor while maintaining low cost and low power consumption.

CLASSIFICATIONS OF MICROPROCESSORS

DIGITAL SIGNAL PROCESSORS (DSP)

- **DSP Architecture Features:**
  - Memory architecture designed for streaming data, using DMA extensively
  - Deliberate exclusion of a memory management unit (MMU)
  - Bit-reversed addressing, a special addressing mode useful for calculating FFTs
  - Special arithmetic operations, such as fast multiply–accumulates (MACs)
  - Separate program and data memories (Harvard architecture)

CLASSIFICATIONS OF MICROPROCESSORS

APPLICATION-SPECIFIC INSTRUCTION SET PROCESSORS (ASIP)

- ASIP is typically a programmable architecture that is designed in a specific way to perform certain tasks more efficiently
- As the name suggests, the Instruction set seems to be the core characteristic of any ASIP based platform; but this is entirely not true.
- Considering a whole platform, other very important attributes like interfaces and micro-architecture do contribute a lot to the overall system performance.

CLASSIFICATIONS OF MICROPROCESSORS

APPLICATION-SPECIFIC INSTRUCTION SET PROCESSORS (ASIP)

- The term "Application" in ASIP is not necessarily related to software applications, it actually describe the class of tasks the ASIP platform was designed to efficiently accomplish
- This extra efficiency is not exclusively associated with faster performance.
- Other factors like reduced production costs, simplified manufacturing process and less power consumption can all be considered efficiency qualities for ASIP.

CLASSIFICATIONS OF MICROPROCESSORS

APPLICATION-SPECIFIC INTEGRATED CIRCUITS (ASIC)

- ASIC is basically an integrated circuit designed and used by a single company in a specific system.
- Example: An IC designed for a specific line of cellular phones of a company, whereby no other company can use it
- They are incredibly expensive, time-consuming, and resource-intensive to develop
- But extremely high performance coupled with low power consumption.

CLASSIFICATIONS OF MICROPROCESSORS

APPLICATION-SPECIFIC INTEGRATED CIRCUITS (ASIC)

- **Full-custom:** entirely tailor-fitted to a particular application from the very start
- **Semi-custom:** designed to allow a certain degree of modification during the manufacturing process.
- **Structured:** built from a group of 'platform slices', with a 'platform slice' being defined as a pre-manufactured device, system, or logic for that platform.
- **Gate-array:** ASIC are transistors which are predefined on the silicon wafer.

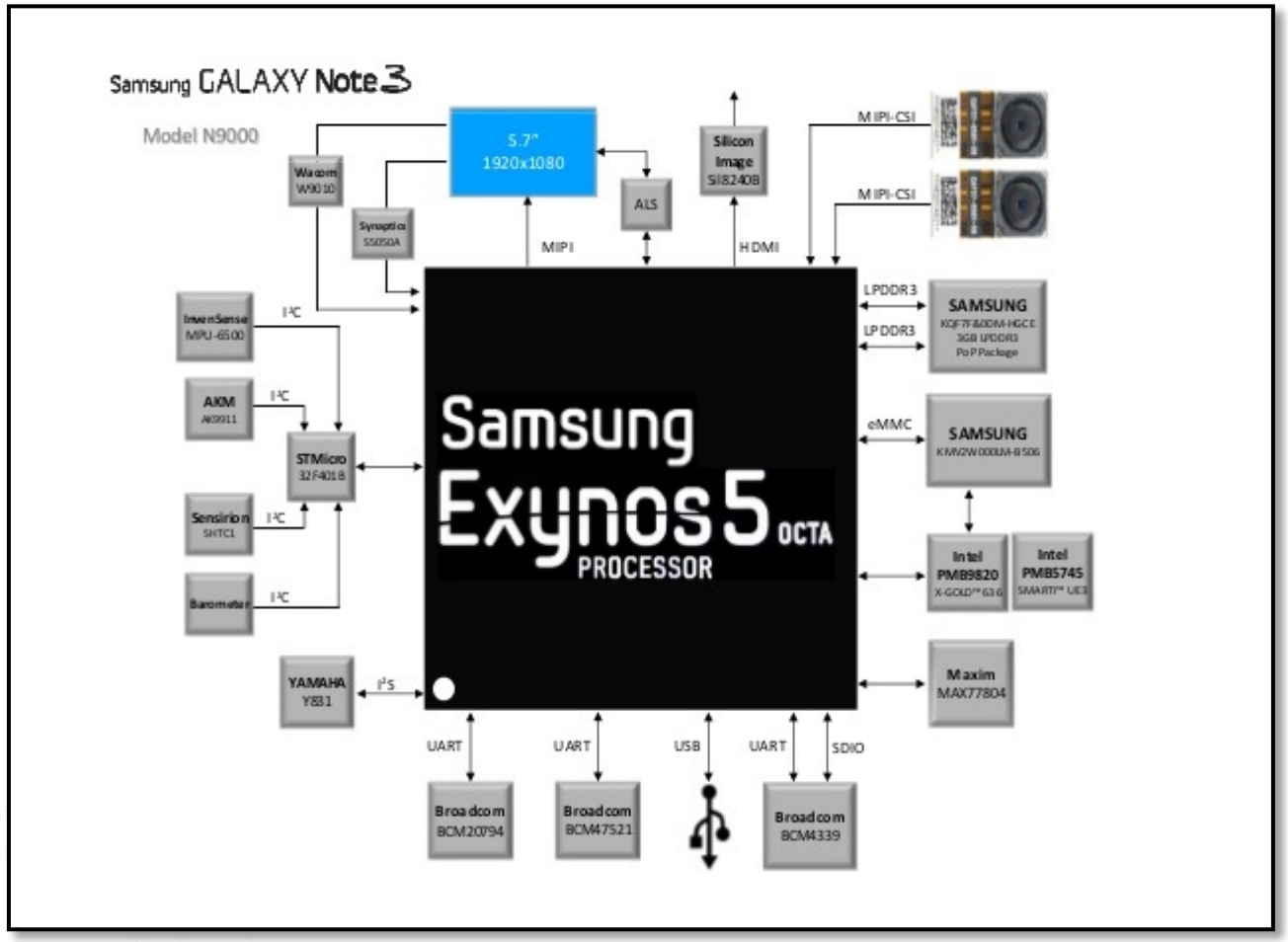
CLASSIFICATIONS OF MICROPROCESSORS

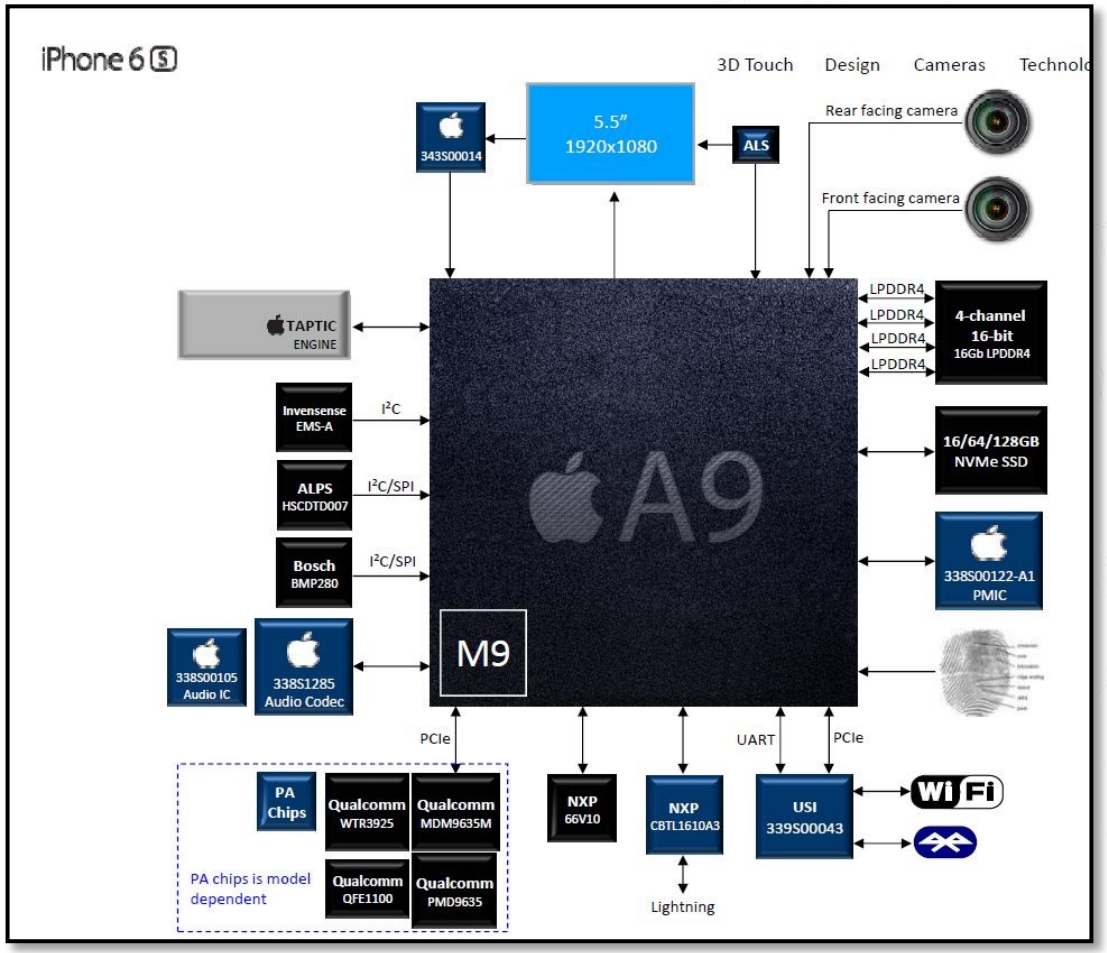
APPLICATION-SPECIFIC INTEGRATED CIRCUITS (ASIC)

- **Applications of ASIC:**
  - An IC that encodes and decodes digital data using a proprietary encoding/decoding algorithm.
  - A medical IC designed to monitor a specific human biometric parameter.
  - An amplifier IC designed to meet certain specifications not available in standard amplifier products.
  - An IC that's custom-made for a particular automated test equipment.

CLASSIFICATIONS OF MICROPROCESSORS

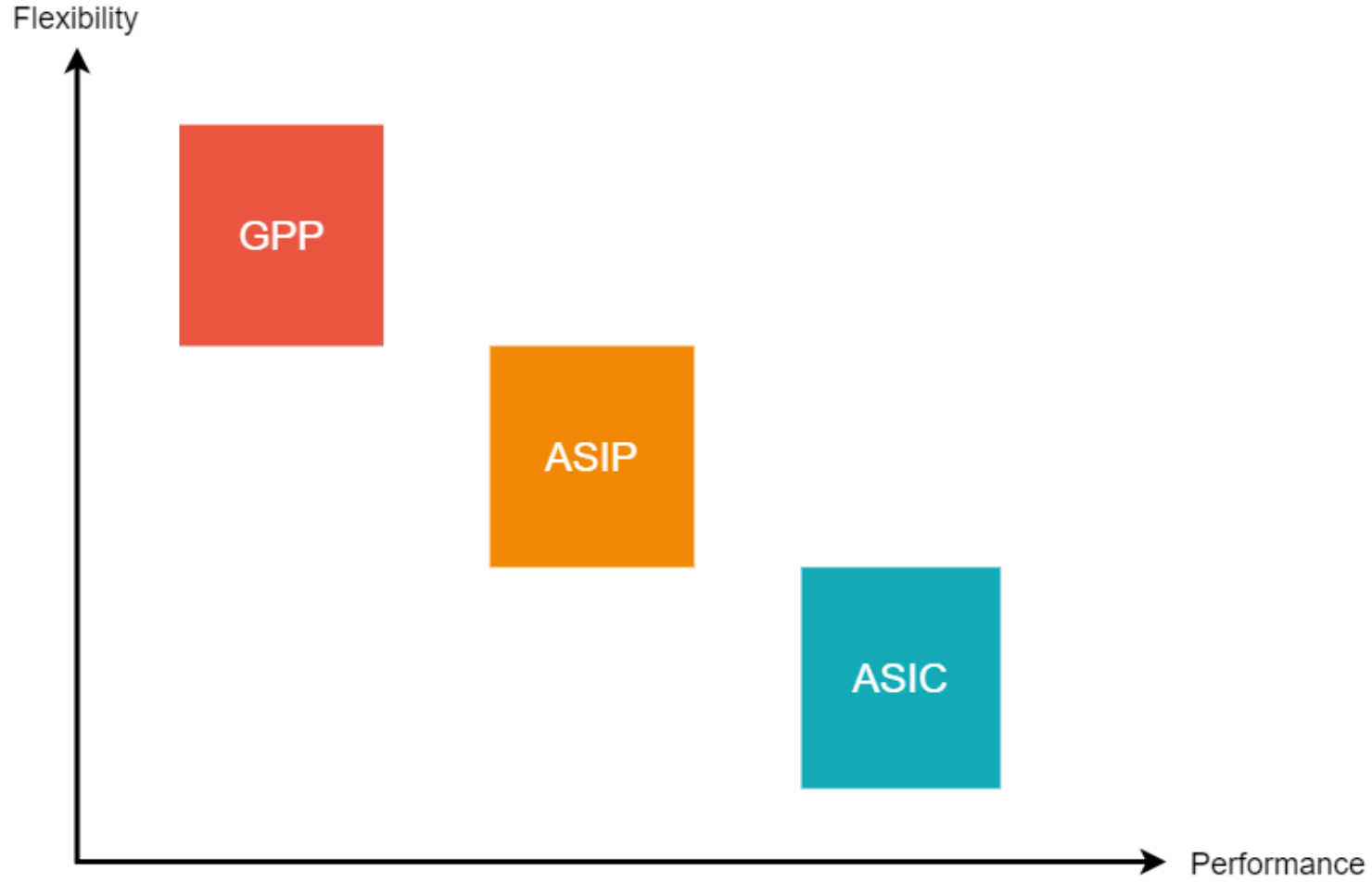
APPLICATION-SPECIFIC INTEGRATED CIRCUITS (ASIC) - EXAMPLE





CLASSIFICATIONS OF MICROPROCESSORS

Compares the Processors Performance vs. Their Flexibility



CLASSIFICATIONS OF MICROPROCESSORS

Different Approaches for Building Embedded System

	GPP	ASIP	ASIC
Performance	Low	High	Very High
Flexibility	Excellent	Good	Poor
HW Design	None	Large	Very Large
SW Design	Small	Large	None
Power	Large	Medium	Small
Reuse	Excellent	Good	Poor
Market	Very Large	Relatively Large	Small
Cost	Mainly on SW	SOC	Volume sensitive

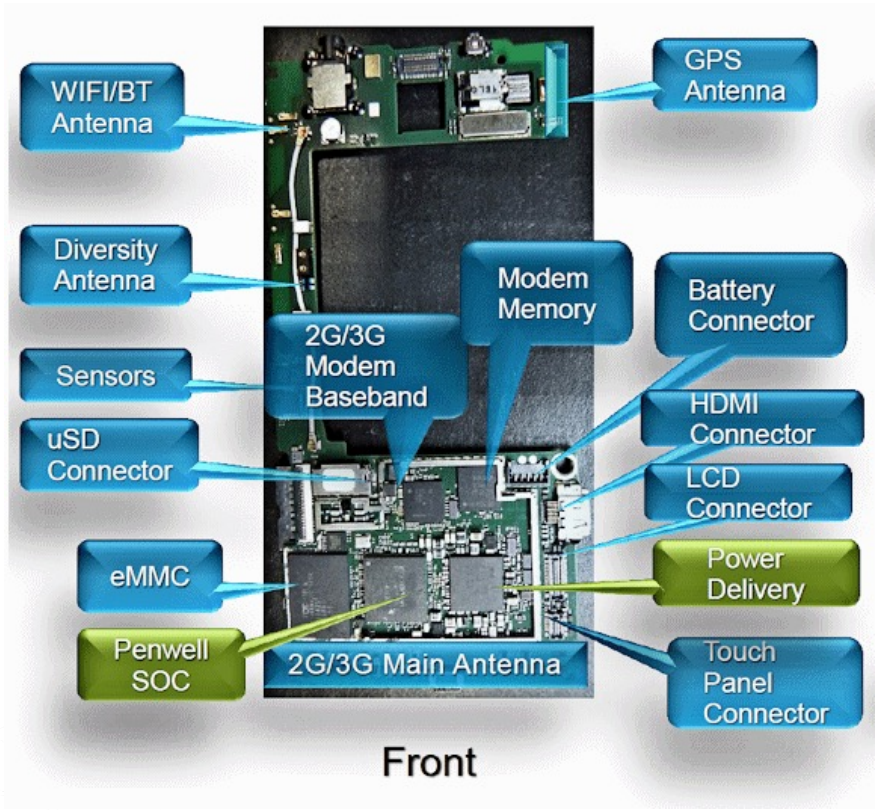
CLASSIFICATIONS OF MICROPROCESSORS

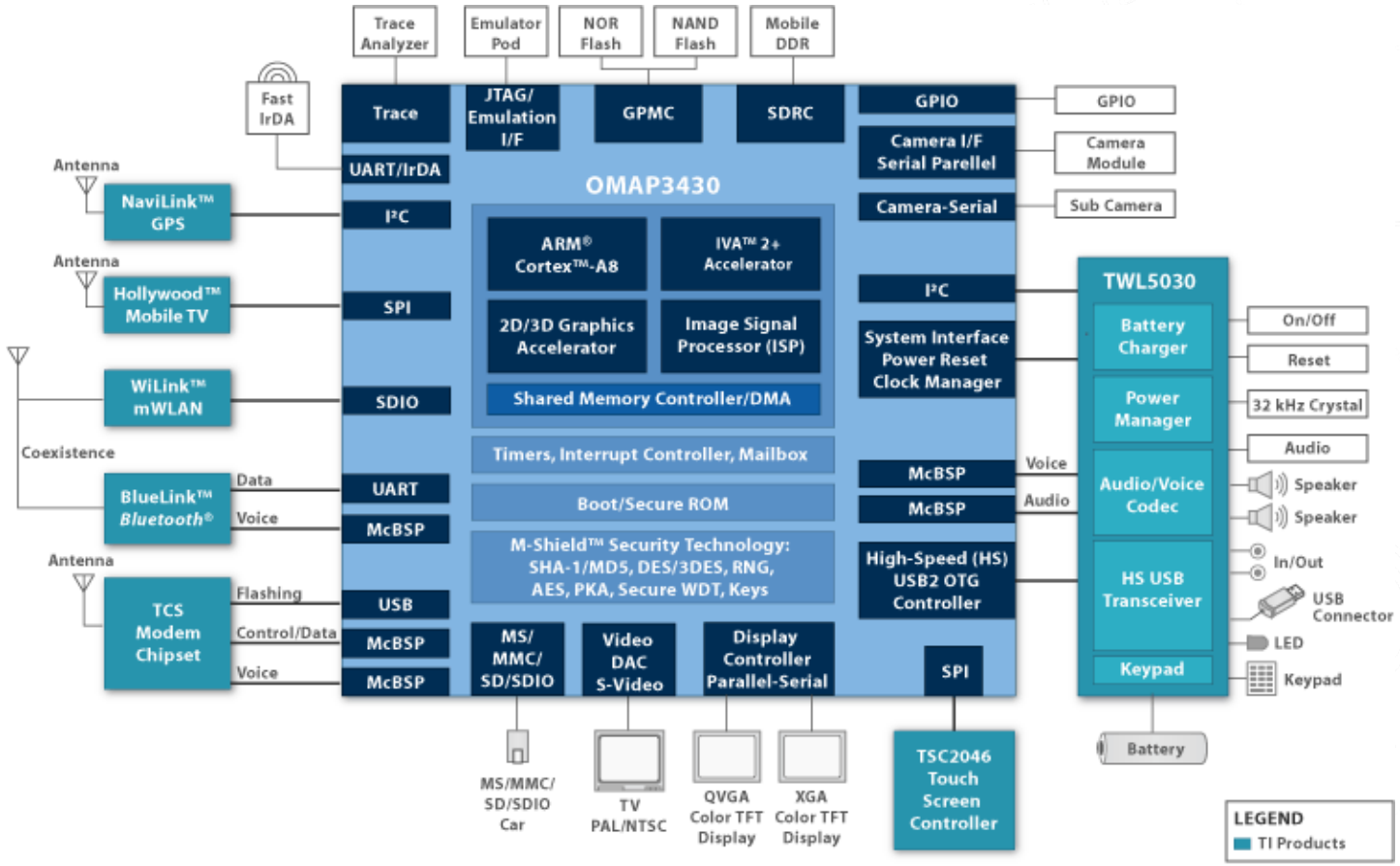
SYSTEM ON CHIP (SoC)

- SoC is an integrated circuit (IC) that takes a single platform and integrates an entire electronic system onto it. It is, exactly as its name suggests, an entire system on a single chip.
- SoC contains one or more processor cores — microprocessors (MPUs) and/or microcontrollers (MCUs) and/or digital signal processors (DSPs) — along with on-chip memory, hardware accelerator functions, peripheral functions etc.,
- SoC can perform a variety of functions including signal processing, wireless communication, artificial intelligence etc.

CLASSIFICATIONS OF MICROPROCESSORS

SYSTEM ON CHIP (SoC) – EXAMPLE

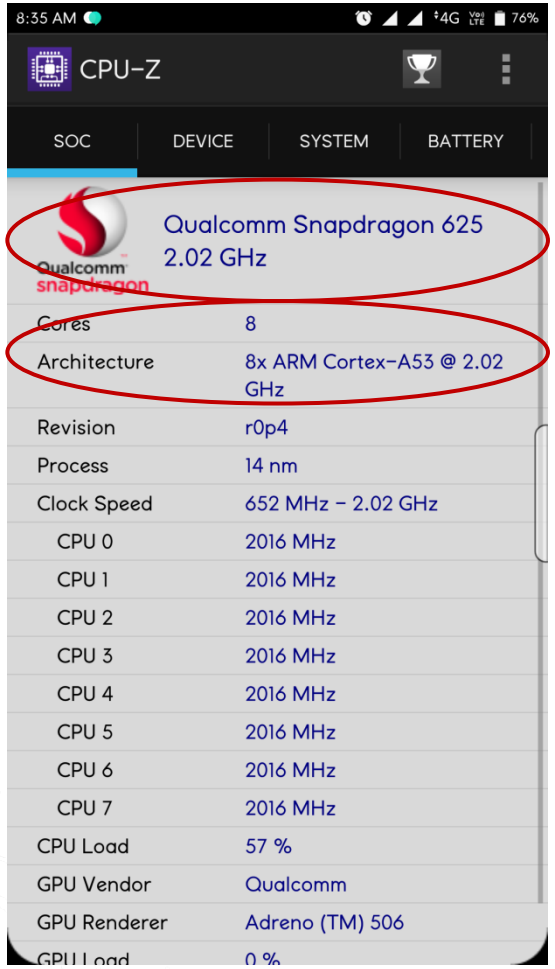


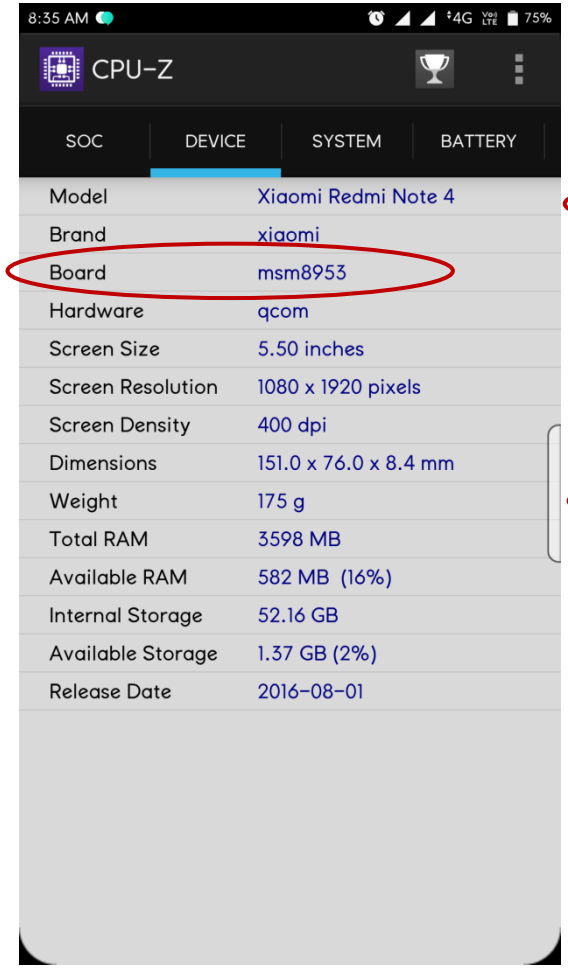


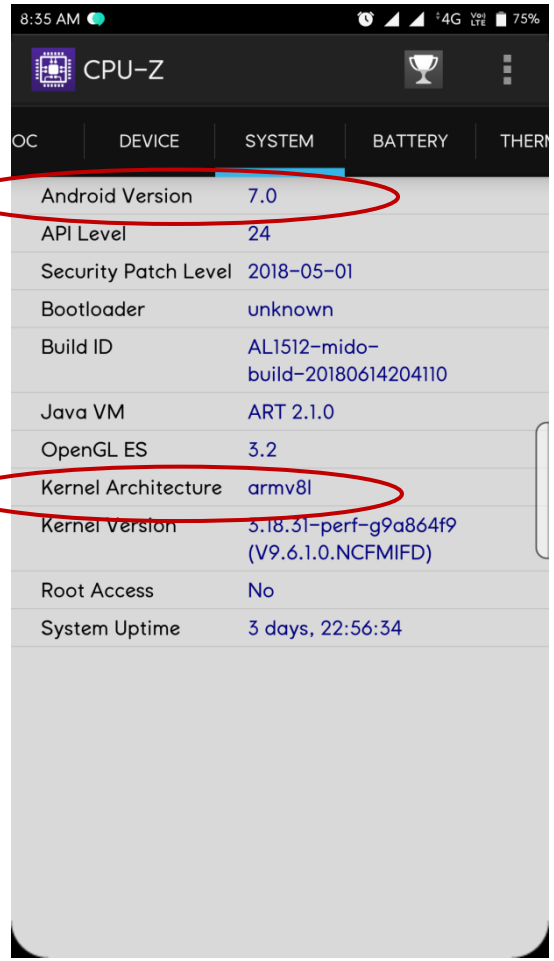
What's Inside a Smartphone? Video

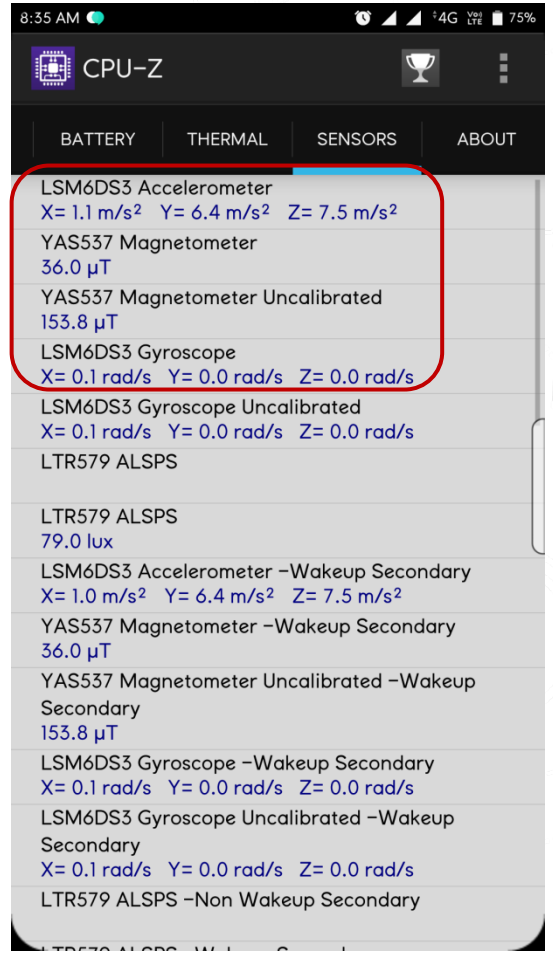
CLASSIFICATIONS OF MICROPROCESSORS

SYSTEM ON CHIP (SoC) – EXAMPLE REDMI NOTE 4 (CPU-Z APP)



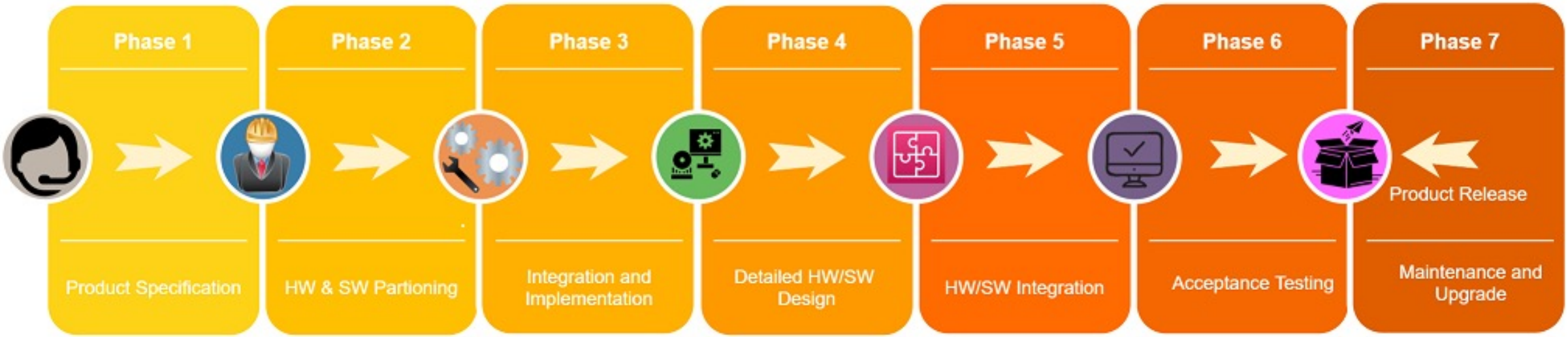






EMBEDDED SYSTEM DESIGN LIFE CYCLE

EMBEDDED SYSTEM DESIGN LIFE CYCLE



Ref. "Embedded Systems Design: An Introduction to Processes, Tools, and Techniques" by Arnold S. Berger

EMBEDDED SYSTEM DESIGN LIFE CYCLE

PHASE-1: PRODUCT SPECIFICATION

- **Objective:** capture a formal description of the complete system
- Transforming the customer's wishes into the final product
- Costumer Research Tour
- Sales and Marketing Department take the lead.
- R&D are not allowed(in general)
- Help to arrive a common Vision of the Product

EMBEDDED SYSTEM DESIGN LIFE CYCLE

PHASE-2: HARDWARE & SOFTWARE PARTITIONING

- **Objective:** To decide which Portion in H/W and which in S/W.
- Example, FPU (H/W) faster, Floating Point (S/W) slower.
- partitioning decision is a complex optimization problem
- Choice of CPU impacts the partitioning decision

EMBEDDED SYSTEM DESIGN LIFE CYCLE

PHASE-3: ITERATION AND IMPLEMENTATION

- **Objective:** Perform initial design before the detailed implementation
- Final stage before H/W and S/W team stop communication
- Hardware designers might be using simulation tools
- Software designers running code on single-board computers that use the target micro processor
- The Design can be extended in H/W or S/W according to convenience.

EMBEDDED SYSTEM DESIGN LIFE CYCLE

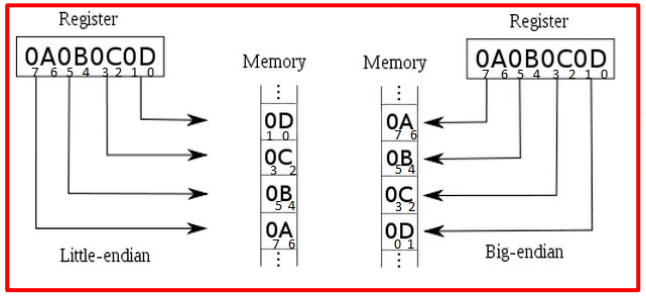
PHASE-4: DETAILED HARDWARE AND SOFTWARE DESIGN

- Objective: Getting the hardware and software done for the next phase
- S/W embedded developer must understand about execution environment, development tools, and run-time package.
- H/W embedded systems developer must understand about,
  - Memory usage.
  - System startup
  - Interrupts and exceptions

EMBEDDED SYSTEM DESIGN LIFE CYCLE

PHASE-5: HARDWARE/SOFTWARE INTEGRATION

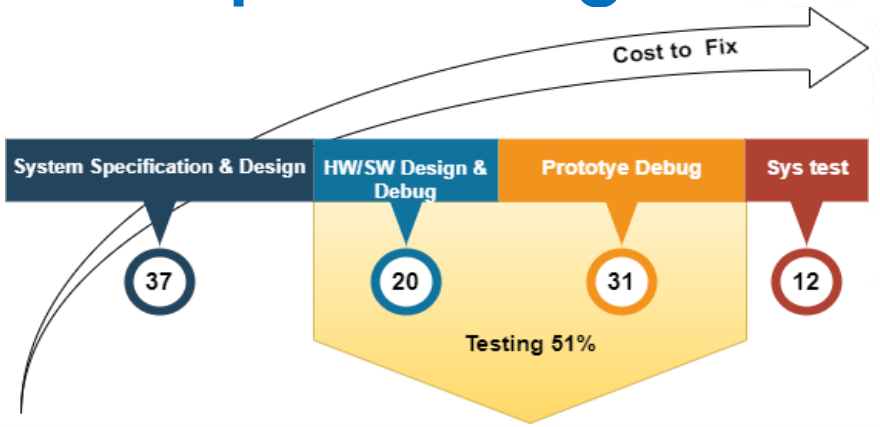
- Objective: To integrate both H/W and S/W to carryout testing
- This stage require special tools and methods to manage the complexity.
- Helps perform debugging and identify errors, Also discover whether S/W team understood the H/W description.
- Example, Big Endian/Little Endian problem



EMBEDDED SYSTEM DESIGN LIFE CYCLE

PHASE-6: PRODUCT TESTING AND RELEASE

- Objective : Verify designed system meet its requirement or not also ensure software doesn't crash at a critical moment
- Testing & reliability requirements-more stringent in embedded
- Testing must determine whether the system is performing close to its optimal capabilities
- Who does the testing? Not the designer



EMBEDDED SYSTEM DESIGN LIFE CYCLE

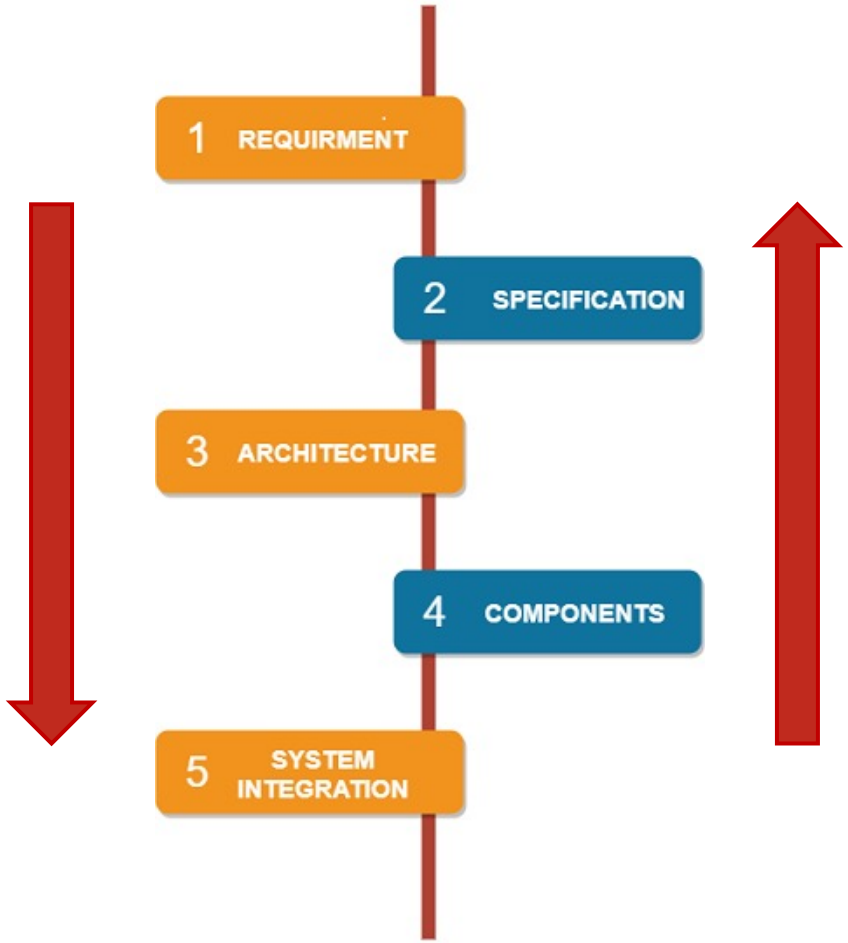
PHASE-7: MAINTAINING AND UPGRADING EXISTING PRODUCTS

- **Objective:** To maintain and add additional features aon the existing product design
- Carried out by non-members of the original design team for a particular product
- Experience, skills, the existing documentation, and the old product to understand the original design well enough to maintain and improve it.

EMBEDDED SYSTEM DESIGN PROCESS

EMBEDDED SYSTEM DESIGN PROCESS

Top–down - From description of the system to concrete details.



bottom–up - Start with components to build a system.

Ref. "Computers as components: principles of embedded computing system design" by Wayne Wolf

EMBEDDED SYSTEM DESIGN PROCESS

- The major goals of the design to be considered are :
  - Manufacturing cost
  - Performance
  - Power consumption
- Tasks which need to be performed at each step,
  - analyze the design at each step - meet specification
  - refine the design - add details
  - verify the design - system goals

EMBEDDED SYSTEM DESIGN PROCESS

REQUIREMENTS

- Informal descriptions gathered from the customer are known as requirements
- Requirements are refined into a specification to begin the designing of the system architecture
- Can be functional or non-functional requirements
  - **Functional:** Needed output as a function of input
  - **Non-functional:** performance, cost, physical size, weight, and power consumption

EMBEDDED SYSTEM DESIGN PROCESS

REQUIREMENTS – EXAMPLE (GPS Navigation System)

Product Name	GPS Navigation System
Purpose	Consumer grade moving map for driving use
Inputs	Power button two control button
Outputs	Back-lit LCD Display 400x600
Functions	Uses 5-receiver GPS system; three user-selectable resolutions; always displays current latitude and longitude
Performance	Updates screen within 0.25 seconds upon movement
Power	100mW
Cost	30 USD
Size And Weight	2" x 6" and 250 grams

EMBEDDED SYSTEM DESIGN PROCESS

SPECIFICATION

- Requirements gathered is refined into a specification.
- Specification serves as the contract between the customers and the architects.
- Specification is essential to create working systems with a minimum of designer effort.
- Specific, understandable and accurately reflect the customer's requirements.

EMBEDDED SYSTEM DESIGN PROCESS

SPECIFICATION - EXAMPLE (GPS Navigation System)

- Specification would include details for several components:
  - Data received from the GPS satellite
  - Map data
  - User interface
  - Operations that must be performed to satisfy customer requests
  - Background actions

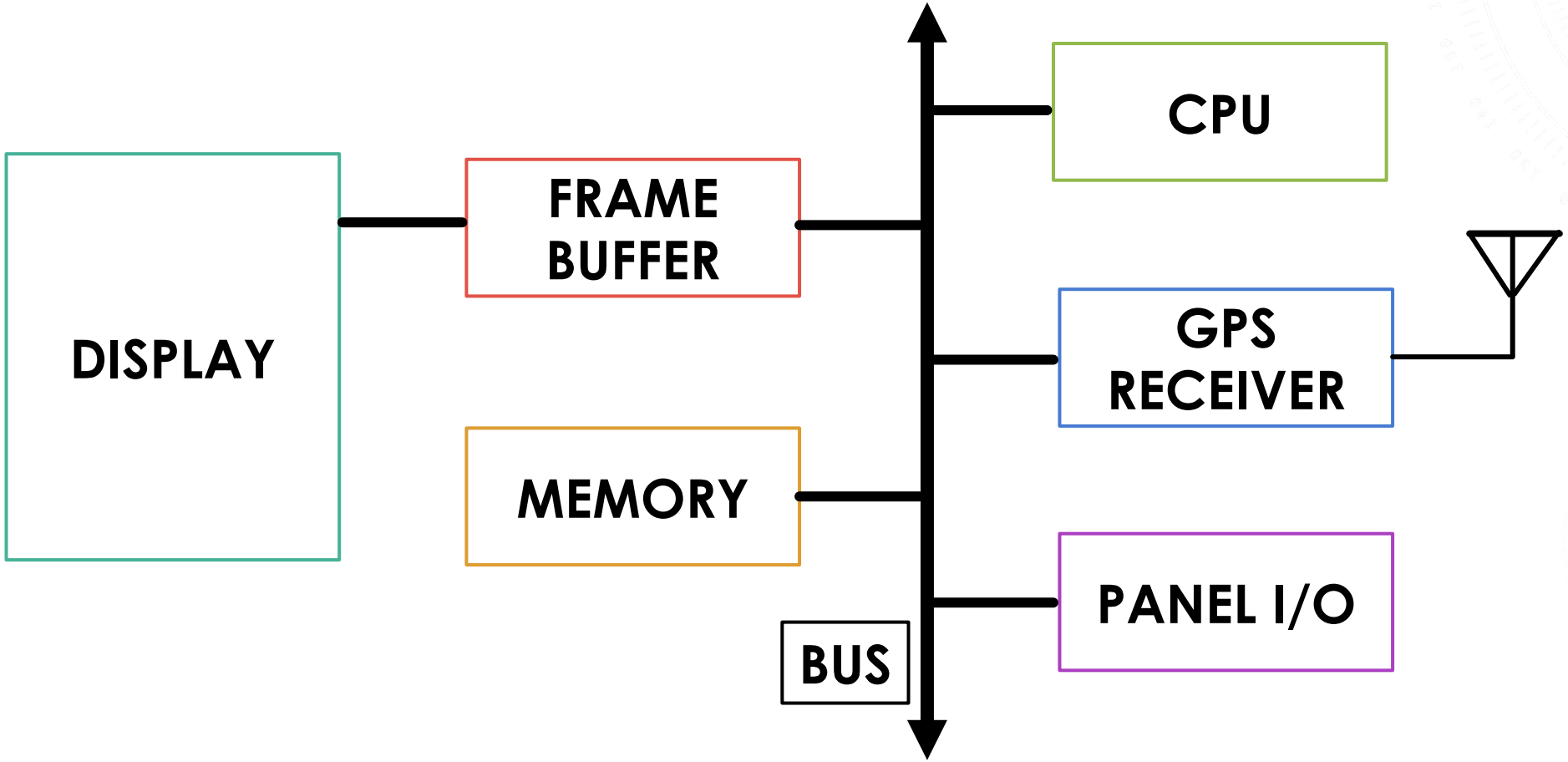
EMBEDDED SYSTEM DESIGN PROCESS

ARCHITECTURE

- The specification describes only the functions of the system
- Implementation of the system is described by the Architecture
- The architecture is a plan for the overall structure of the system.
- It will be used later to design the components
- The system block diagram may be refined into two block diagrams - hardware and software

EMBEDDED SYSTEM DESIGN PROCESS

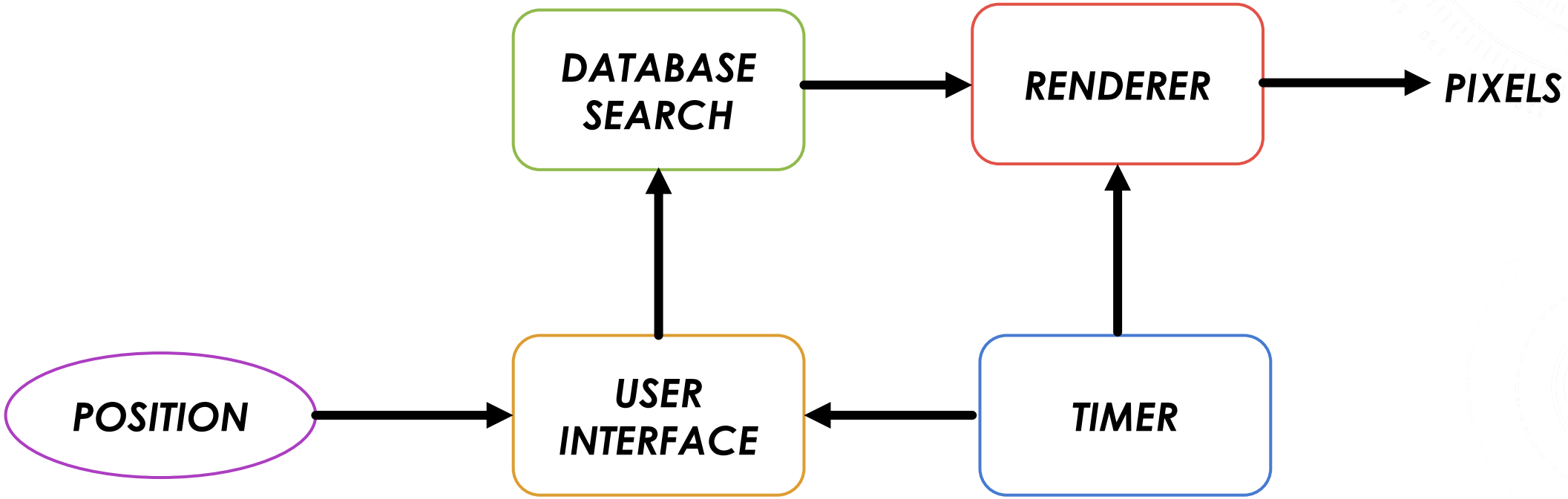
ARCHITECTURE - EXAMPLE (GPS Navigation System)



SYSTEM DESIGN - HARDWARE

EMBEDDED SYSTEM DESIGN PROCESS

ARCHITECTURE - EXAMPLE (GPS Navigation System)



SYSTEM DESIGN - SOFTWARE

EMBEDDED SYSTEM DESIGN PROCESS

HARDWARE AND SOFTWARE COMPONENTS

- The architectural description specify what components we need
- The component design effort builds those components in conformance to the architecture and specification
- Includes both hardware and software modules
- Some of the components will be ready-made (example :CPU, memory chips)

EMBEDDED SYSTEM DESIGN PROCESS

HARDWARE AND SOFTWARE COMPONENTS - EXAMPLE

- In the moving map, GPS receiver is a predesigned standard hardware component.
- Topographic software is a standard software module which uses standard routines to access the database.
- Printed circuit board are the components which needs to be designed.
- When creating these embedded software modules, ensure the system runs properly in real time and that it does not take up more memory space than allowed.

EMBEDDED SYSTEM DESIGN PROCESS

SYSTEM INTEGRATION

- After the components are built, they are integrated
- Bugs are typically found during the system integration
- By debugging a few modules at a time, simple bugs can be discovered
- By fixing the simple bugs early, more complex bugs can be discovered
- Appropriate debugging facilities during design which can help to ease system integration problems.

EMBEDDED SYSTEM DESIGN PROCESS – EXAMPLE

AUTOMATIC CHOCOLATE VENDING MACHINE

REQUIREMENTS

- **Purpose:**
  - To sell chocolate through an ACVM from which children can automatically purchase the chocolates.
  - The payment is by inserting the coins into a coin-slot.
- **Inputs:**
  - Coins of different denominations through a coin slot.
  - User commands.
- **Outputs:**
  - Chocolate and signal to the system that subtracts the cost from the value of amount collected.
  - Display of the menus for GUIs, time and date, advertisements, welcome and thank messages.

AUTOMATIC CHOCOLATE VENDING MACHINE

REQUIREMENTS

- **Functions of the system:**
  - A child sends commands to the system using a GUI. It must consists of the LCD display and keypad units.
  - The child inserts the coins for cost of chocolate and the machine delivers the chocolate.
  - If the coins are not inserted as per the cost of chocolate in reasonable times then all coins are refunded. If the coin amount more, the excess amount is refunded along with chocolate.
  - The coins for the chocolates purchased collect inside the machine in a collector channel, so that owner can get the money, again through appropriate commands using the GUI.
  - USB wireless modem enables communication through Internet to the ACVM owner.

AUTOMATIC CHOCOLATE VENDING MACHINE

REQUIREMENTS

- **Design metrics:**
  - **Power Dissipation:** As required by mechanical units, display units and computer system
  - **Performance:** One chocolate in two minutes and 256 chocolates before next filling of chocolates into the machine.[Assumed]
  - **Process Deadlines:** Machine waits for maximum 30 s for the coins and machine should deliver the chocolate within 60 s.
  - **User Interfaces:** Graphic at LCD or touch screen display on LCD and commands by children or machine owner through fingers on keypad or touch screen
  - **Engineering Cost:** US$ 50000 (assumed)
  - **Manufacturing Cost:** US$ 1000 (assumed)

AUTOMATIC CHOCOLATE VENDING MACHINE

SPECIFICATIONS

- **Alphanumeric keypad** on the top of the machine. A child interaction with it when buying a chocolate. Owner commands and interaction with the machine.

- **Three line LCD display** unit on the top of the machine. Displays menus, entered text, pictograms, and welcome, thank and other messages, and time and date. Child as well as the ACVM owner GUIs with the machine using keypad and display.

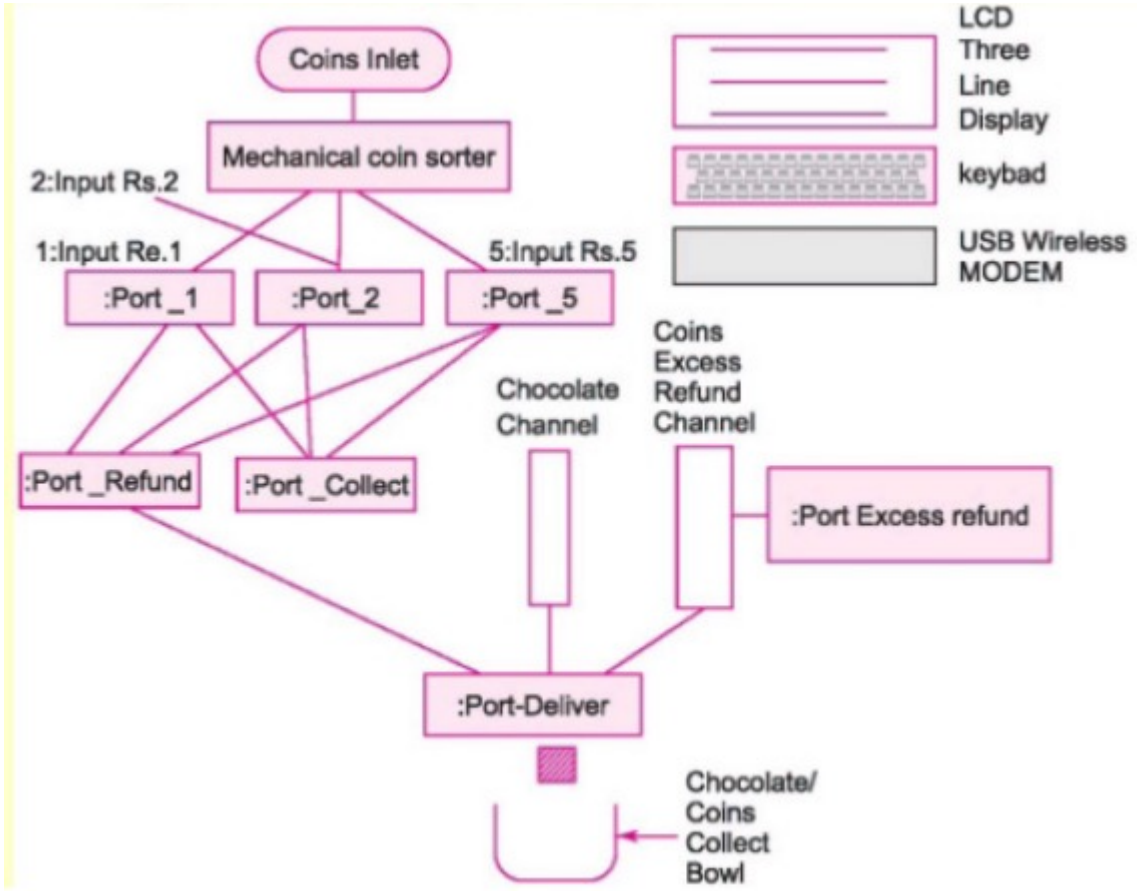
AUTOMATIC CHOCOLATE VENDING MACHINE

SPECIFICATIONS

- **Coin insertion slot** so that the child can insert the coins to buy a chocolate.
- **Delivery slot** to collect the chocolate, and coins if refunded.
- **Internet connection port** so that owner can interact with ACVM from remote.

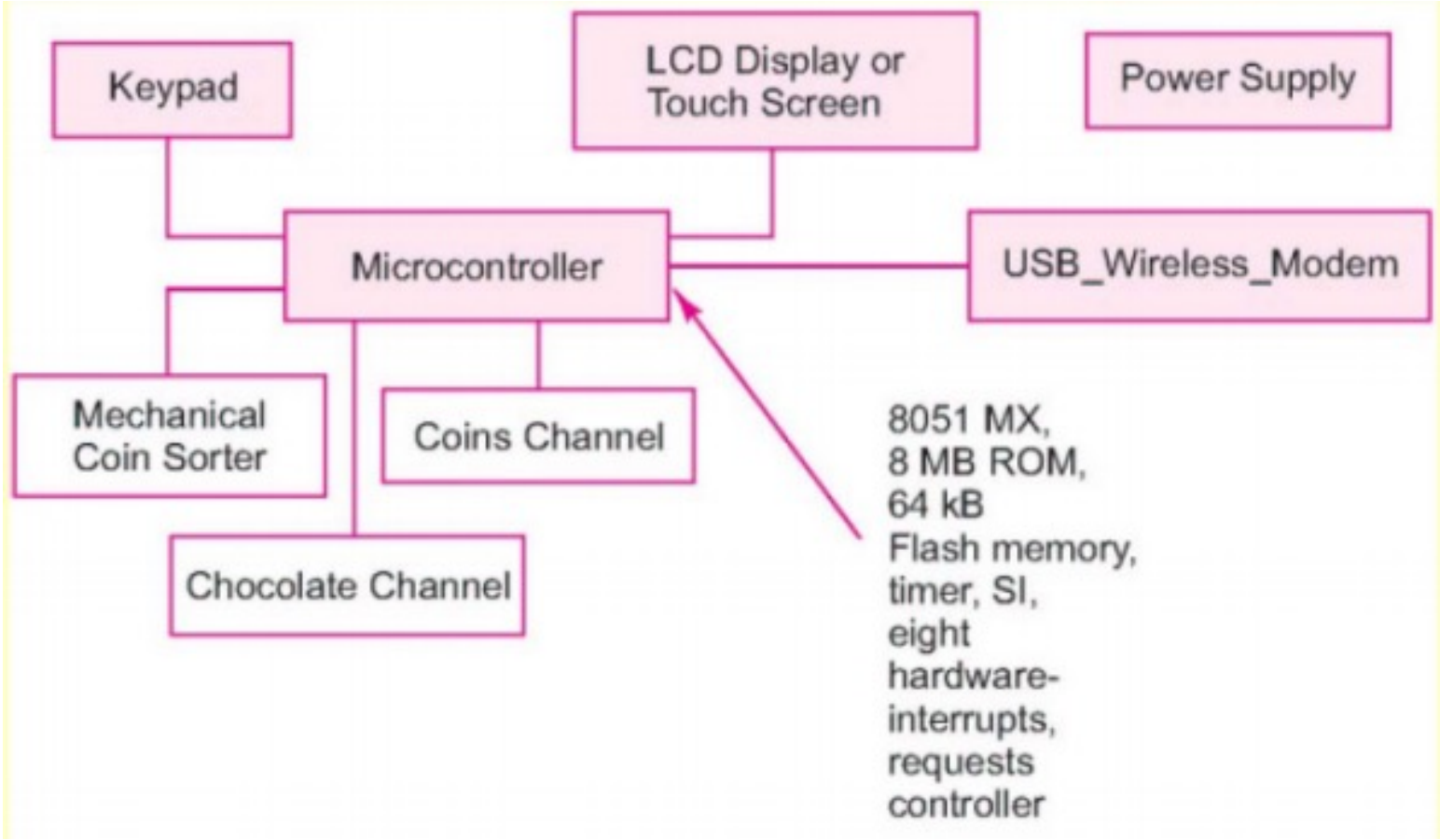
AUTOMATIC CHOCOLATE VENDING MACHINE

ARCHITECTURE – OVERALL SYSTEM DIAGRAM



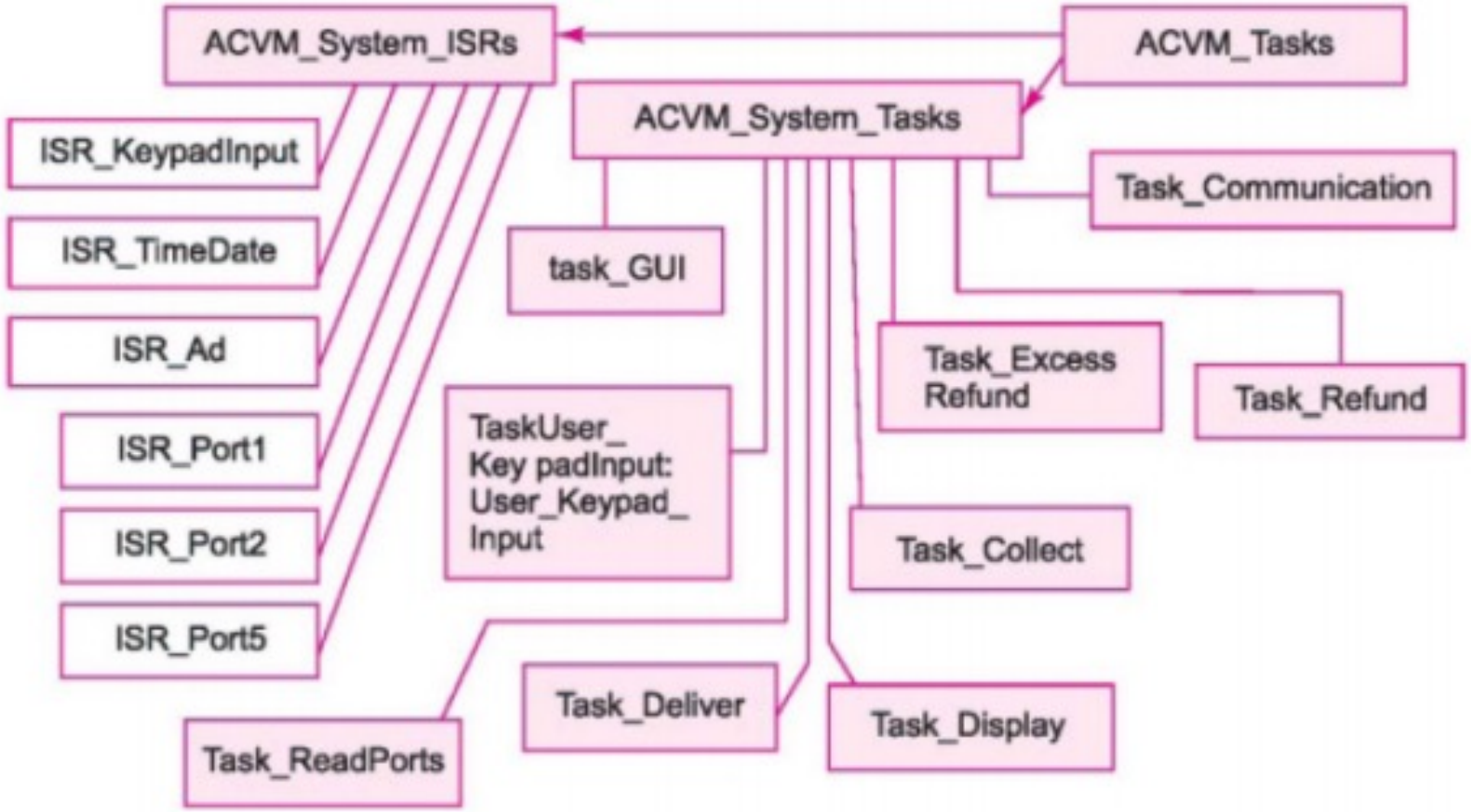
AUTOMATIC CHOCOLATE VENDING MACHINE

ARCHITECTURE – HARDWARE DIAGRAM



AUTOMATIC CHOCOLATE VENDING MACHINE

ARCHITECTURE – SOFTWARE DIAGRAM



AUTOMATIC CHOCOLATE VENDING MACHINE

COMPONENTS - HARDWARE

- ACVM specific hardware to sort the coins of different denomination and each denomination.
- Main Power supply 220 V 50 Hz or 110 V 60 Hz. Internal circuits drive by supply of 5 V 50mA for electronic and 12 V 2 A for mechanical systems.
- A TCP/IP port.
- A 1s resolution timer is obtained by programming 8051 timer.
- Flash memory part of ROM and RAM for storing temporary variables and stack.
- Microcontroller 8051 MX and 8 MB ROM for application codes.

AUTOMATIC CHOCOLATE VENDING MACHINE

COMPONENTS - SOFTWARE

- **Task_ReadPorts:** Waits for the coins and action as per coins collected.
- **Task_Collect:** Waits for coins = or > cost till timeout and act accordingly.
- **Task_Deliver:** Waits for Task_Readports, delivers chocolate, and decreases coins' amount after delivery.
- **Task_Refund:** Waits for refund event and calculate refund amount.
- **Task_Excess Refund:** Refunds the Excess amount
- **Task_Display:** Waits for the message and display as per message.

AUTOMATIC CHOCOLATE VENDING MACHINE

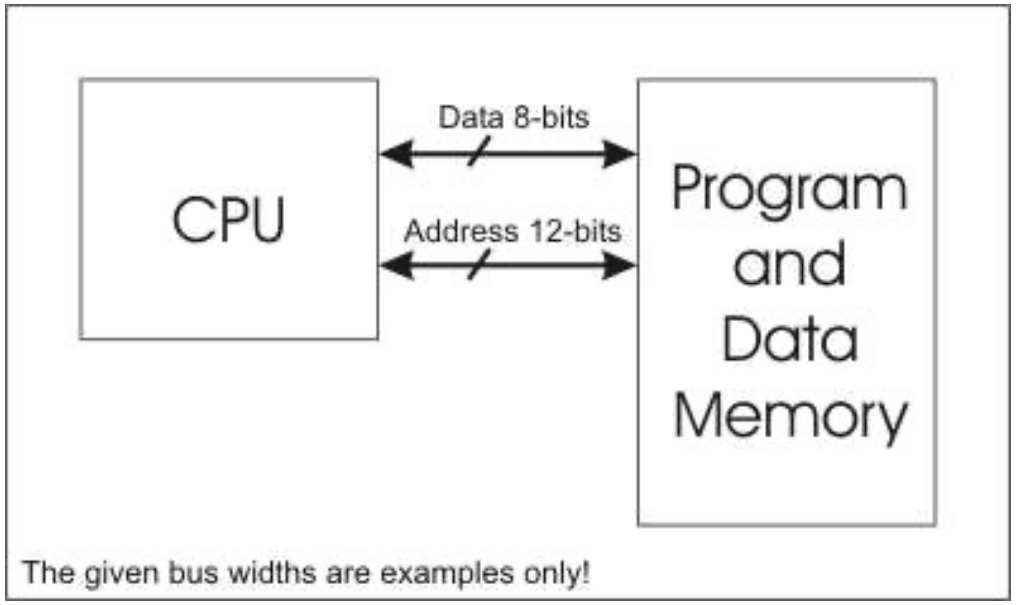
SYSTEM INTEGRATION

- **Test and validation conditions:**
  - All user commands must function correctly.
  - All graphic displays and menus should appear as per the program.
  - Each task should be tested with test inputs.
  - Tested for 60 users per hour.

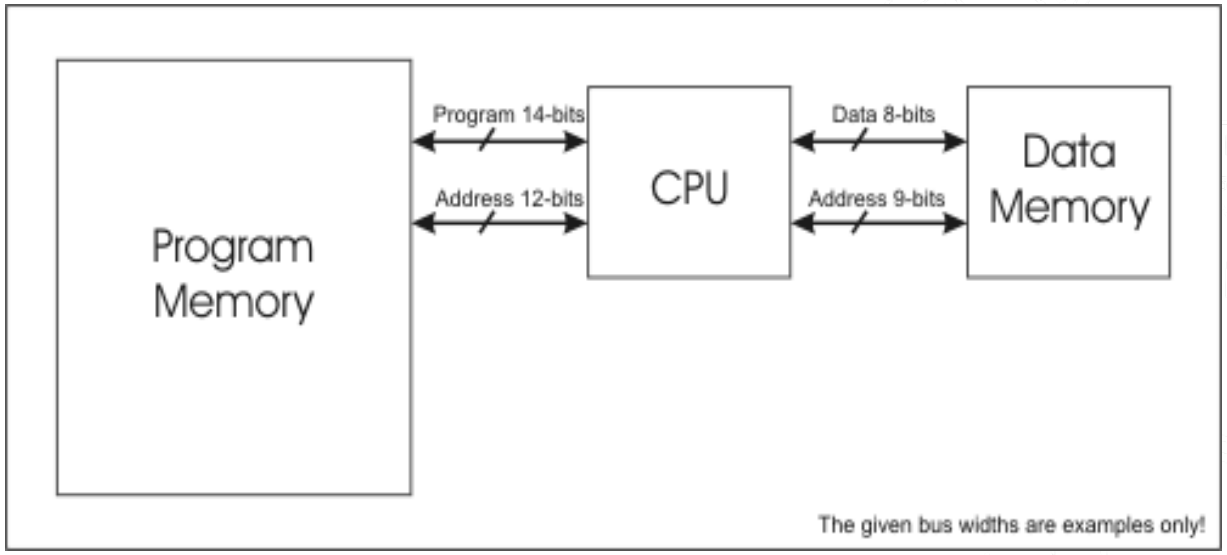
MICROPROCESSOR vs MICROCONTROLLER

Parameters	μP	μC
Architecture	Von-Neumann and RISC architecture	Harvard and CISC architectures.
Flexibility	designer can decide on the amount of ROM, RAM and I/O ports.	fixed amount of on-chip ROM, RAM, I/O ports
Processing Power	High	Low
Design Time	More	Less
Power	High	Low
Protection	Less	More
Cost	High	Low
Examples	8085, 8086	8051

ARCHITECTURE



Von Neumann Architecture



Harvard Architecture

ARCHITECTURE

CISC	RISC
Emphasis on hardware	Emphasis on software
Multiple instruction sizes and formats	Instructions of same set with few formats
Less registers	Uses more registers
More addressing modes	Fewer addressing modes
Extensive use of microprogramming	Complexity in compiler
Instructions take a varying amount of cycle time	Instructions take one cycle time
Pipelining is difficult	Pipelining is easy

X86 Architecture

ARM, Power PC

INTRODUCTION TO INTEL MCS 51 µC

INTEL MCS 51 MICROCONTROLLER

- The Intel MCS-51 (commonly termed 8051) is a Harvard architecture, complex instruction set computing (CISC) Architecture, single chip microcontroller (μC) series developed by Intel in 1980 for use in embedded systems
- Intel's original MCS-51 family was developed using N-type metal-oxide-semiconductor (NMOS) technology
- But later versions, identified by a letter C in their name (e.g., 80C51) used complementary metal-oxide-semiconductor (CMOS) technology and consume less power

INTEL MCS 51 MICROCONTROLLER

FEATURES

- 8-bit CPU
- 64K bytes on-chip program memory (ROM)
- 128 bytes on-chip data memory (RAM)
- 32 I/O pins arranged as four 8-bit ports (P0 - P3)
- 32 general purpose registers each of 8-bit
- Special Function Registers (SFRs) of 128 bytes
- 16-bit Program Counter
- 8-bit Processor Status Word (PSW) & Stack Pointer
- Two 16-bit timer/counters : T0 and T1
- Two external and three internal vectored interrupts
- One full duplex serial I/O (UART)

8051 FAMILY SERIES

Feature	8051	8052	8031
ROM(bytes)	4K	8K	0K
RAM(bytes)	128	256	128
Timers	2	3	2
I/O pins	32	32	32
Serial port	1	1	1
Interrupt sources	6	8	6

8051 FAMILY SERIES

- Although the 8051 is the most popular member of the 8051 family, you will not see "8051" in the part number.
- This is because the 8051 is available in different memory types, such as UV-EPROM, flash, and NV-RAM, all of which have different part numbers.
- The UV-EPROM version of the 8051 is the 8751. The flash ROM version is marketed by many companies including Atmel Corp. and Dallas Semiconductor.

8051 FAMILY SERIES

- The Atmel Flash 8051 is called AT89C51, while Dallas Semiconductor calls theirs DS89C4xO (DS89C420/430/440).
- The NV-RAM version of the 8051 made by Dallas Semiconductor is called DS5000.
- There is also an OTP (one-time programmable) version of the 8051 made by various manufacturers.

8051 MAJOR MANUFACTURERS

- ATMEL
- ANALOG DEVICES
- ST MICROELECTRONICS
- DALLAS
- MAXIM
- SILICON LABS
- TEXAS INSTRUMENTS
- MICROCHIP
- ZILOG

INTEL MCS 51 MICROCONTROLLER

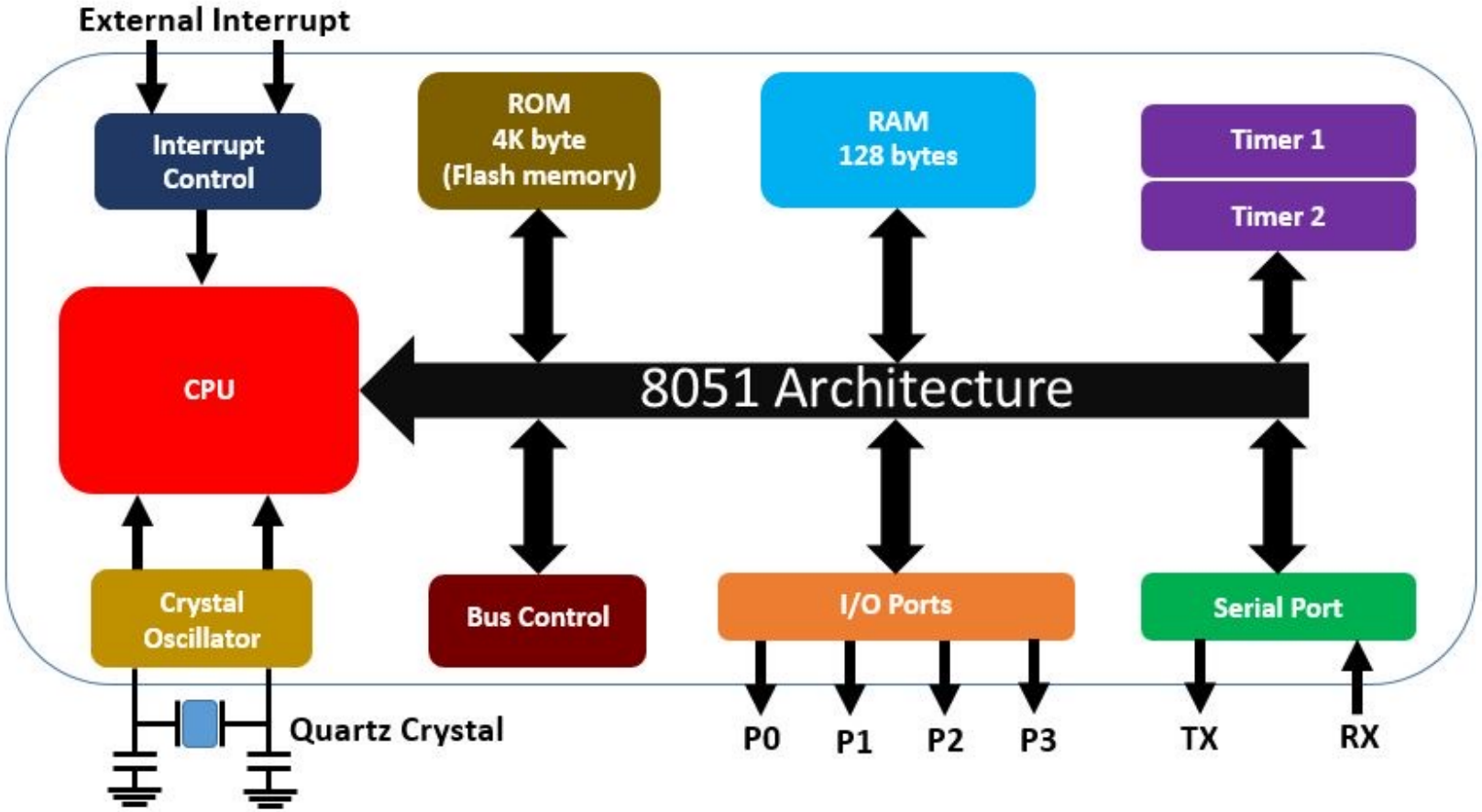
8051 ATMEL SERIES

Part Number	ROM	RAM	I/O pins	Timer	Interrupt	Vcc	Packaging
AT89C51	4K	128	32	2	6	5V	40
AT89C52	8K	256	32	3	8	5V	40
AT89C1051	1K	64	15	1	3	3V	20
AT89C2051	2K	128	32	3	8	3V	20
AT89LV51	4K	128	32	2	6	3V	40
AT89LV52	8K	128	32	3	8	3V	40

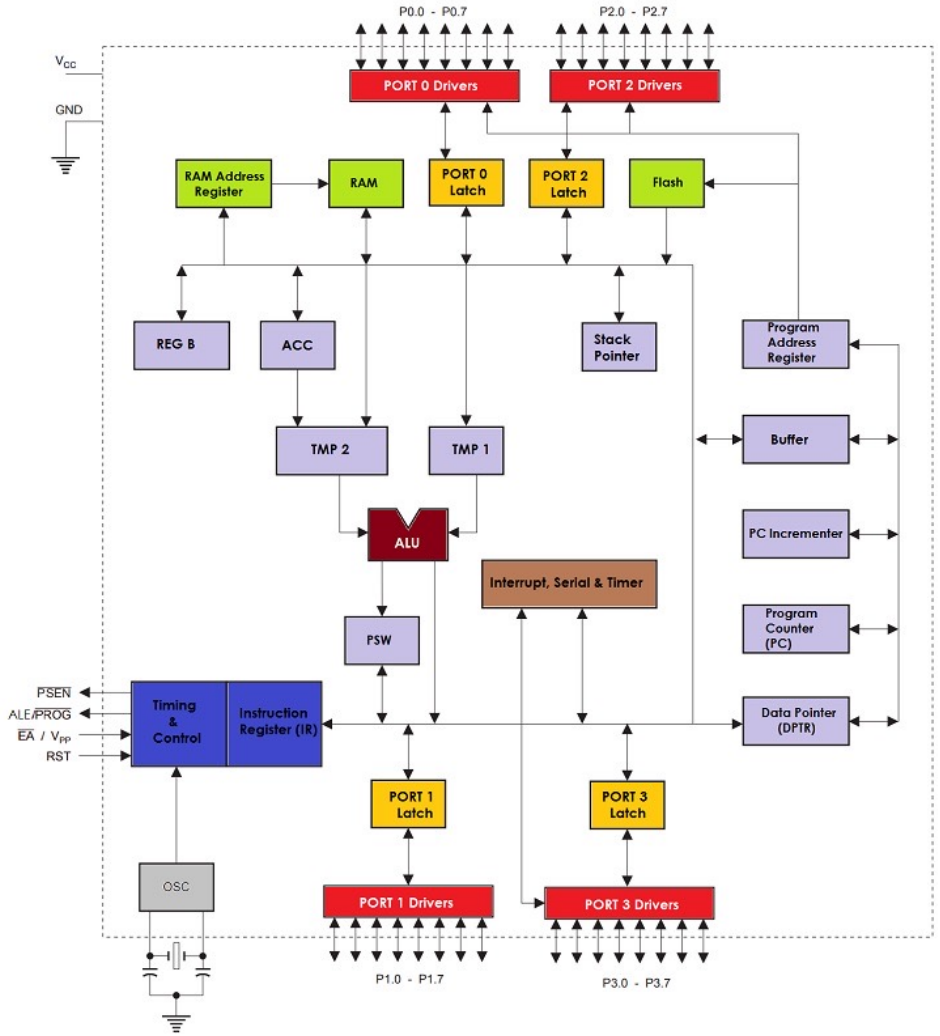
8051 MICROCONTROLLER ARCHITECTURE

8051 MICROCONTROLLER ARCHITECTURE

8051 ARCHITECTURE (Simplified)



8051 MICROCONTROLLER ARCHITECTURE



8051 ARCHITECTURE (Detailed)

8051 MICROCONTROLLER ARCHITECTURE

CENTRAL PROCESSING UNIT (CPU)

- Central Processor Unit (CPU) is the brain of any processing device of the microcontroller.
- It monitors and controls all operations that are performed on the Microcontroller units.
- The User has no control over the work of the CPU directly.
- It reads program written in ROM memory and executes them and do the expected task of that application.

INTERRUPT CONTROL

- There are five interrupt sources in 8051 Microcontroller and interrupt control section control all these interrupts.
- Two external interrupts (INT0 & INT1), two timer (TF0 & TF1) interrupts and one serial port (RI / TI) interrupt.
- The Microcontroller 8051 can be configured in such a way that it temporarily terminates or pause the main program at the occurrence of interrupt. When subroutine is completed then the execution of main program starts as usual.

8051 MICROCONTROLLER ARCHITECTURE

RAM & ROM

- Microcontroller 8051 has 4K of Code Memory or Program memory that is it has 4KB ROM and RAM of 128 bytes.
- The memory which is used to store the program of Microcontroller, is known as code memory or Program memory . It is known as 'ROM'(Read Only Memory).
- Microcontroller also requires a memory to store data or operands temporarily. This memory is known as Data Memory and we use 'RAM'(Random Access Memory) for this purpose.

8051 MICROCONTROLLER ARCHITECTURE

BUS CONTROL

- Bus control section of 8051 is responsible for controlling the operation of address and data bus.
- Bus: Basically Bus is a collection of wires which work as a communication channel or medium for transfer of Data.
- Buses are of two types:
  - Address Bus: Microcontroller 8051 has a 16 bit address bus. It used to address memory locations.
  - Data Bus: Microcontroller 8051 has 8 bits data bus. It is used to carry data.

CRYSTAL OSCILLATOR

- **Crystal Oscillator:** Since Microcontroller is a digital circuit device, therefore it requires clock for its operation.
- For this purpose, Microcontroller 8051 has oscillator circuitry section which works as a clock source for Central Processing Unit.
- As the output pulses of oscillator are stable therefore it enables synchronized work of all parts of 8051 Microcontroller.

8051 MICROCONTROLLER ARCHITECTURE

I/O PORTS

- **I/O Ports:** To connect any external devices or peripherals we require I/O interfacing ports in the microcontroller.
- All 8051 microcontrollers have 4 I/O ports each comprising 8 bits which can be configured as input (1) or an output (0), depends on its logic state.
- Accordingly, in total of 32 input/output pins enabling the microcontroller to be connected to peripheral devices are available for use.

8051 MICROCONTROLLER ARCHITECTURE

TIMERS/COUNTERS

- **Timers/Counters**: 8051 Microcontroller 8051 two 16-bit timers and counters: Timer 0 and Timer 1.
- They can be used either as timers to generate a time delay or as counters to count events happening outside the microcontroller.
- Since the 8051 has an 8-bit architecture, each 16-bit is accessed as two separate registers of low byte and high byte.

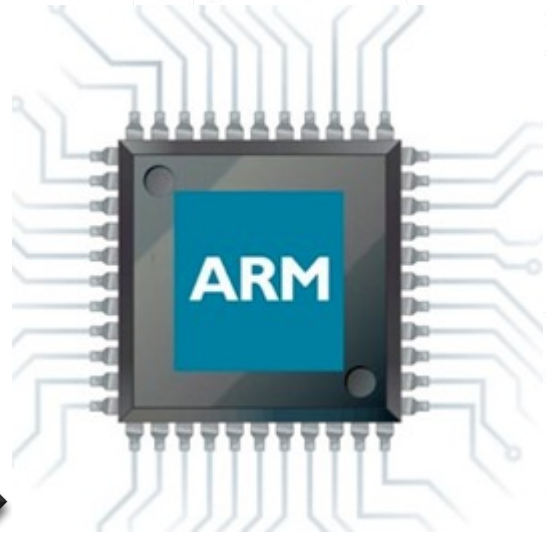
SERIAL PORT

- **Serial port**: The 8051 contains one Serial port or UART (Universal Asynchronous Receiver Transmitter)
- The serial port is full-duplex so, it can transmit and receive simultaneously
- Two port pins are used to provide the serial interface P3.0 is the receive pin (RXD) P3.1 is the transmit pin (TXD)
- This serial port that can be programmed to operate in one of four different modes and at a range of frequencies.

INTRODUCTION TO ARM7

ABOUT ARM

ARM was founded as Advanced RISC Machines in 1990



ARM is the world's leading provider of RISC (Reduced Instruction Set Computer) based microprocessor

ARM does not manufacture silicon by itself, instead it license ARM core design to many semiconductor partners around the world.

Used especially in portable devices due to low power consumption and reasonable performance (MIPS/watt)

ARM also develop technologies to assist designing of ARM architecture which includes, software tools, boards, debug hardware, application software, bus architectures, peripherals etc.,

ARM PROCESSOR - FEATURES

32/64-bit RISC-processor core

Load and Store type architecture

Fast interrupt response

uniform and fixed length instruction set (32-bit ARM & 16-bit Thumb)

small core size and power-efficiency

Multiple stage Pipelined operation

wide range of clock frequency ranging from 1MHz to few GHz

High performance and High code density

8 / 16 / 32/ 64 -bit data types

Various modes of operation (USR, SYS, FIQ,IRQ, SVC, ABT, UND)

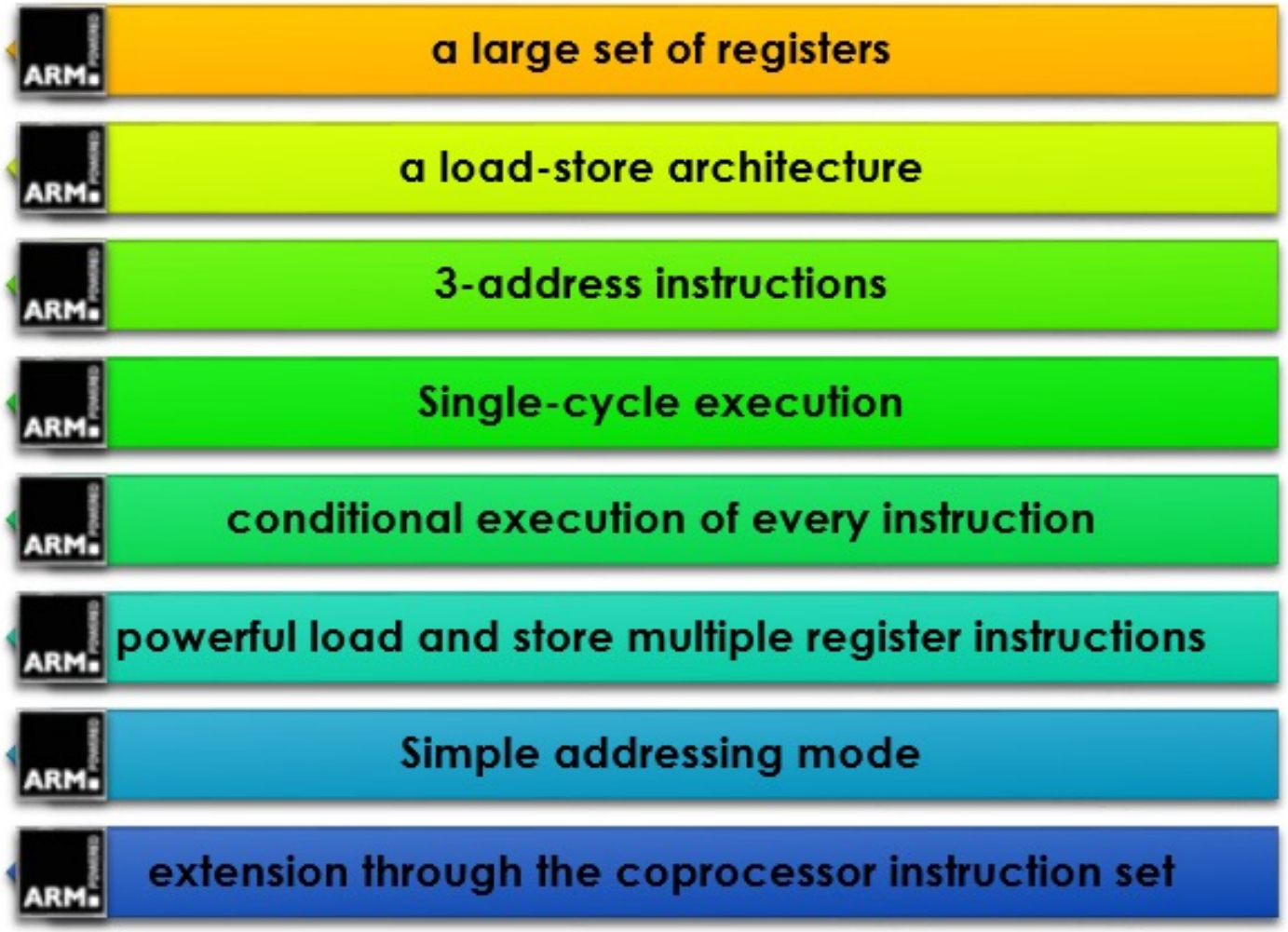
Good speed / power consumption ratio

HIGH PERFORMANCE, LOWER COST, LESS POWER

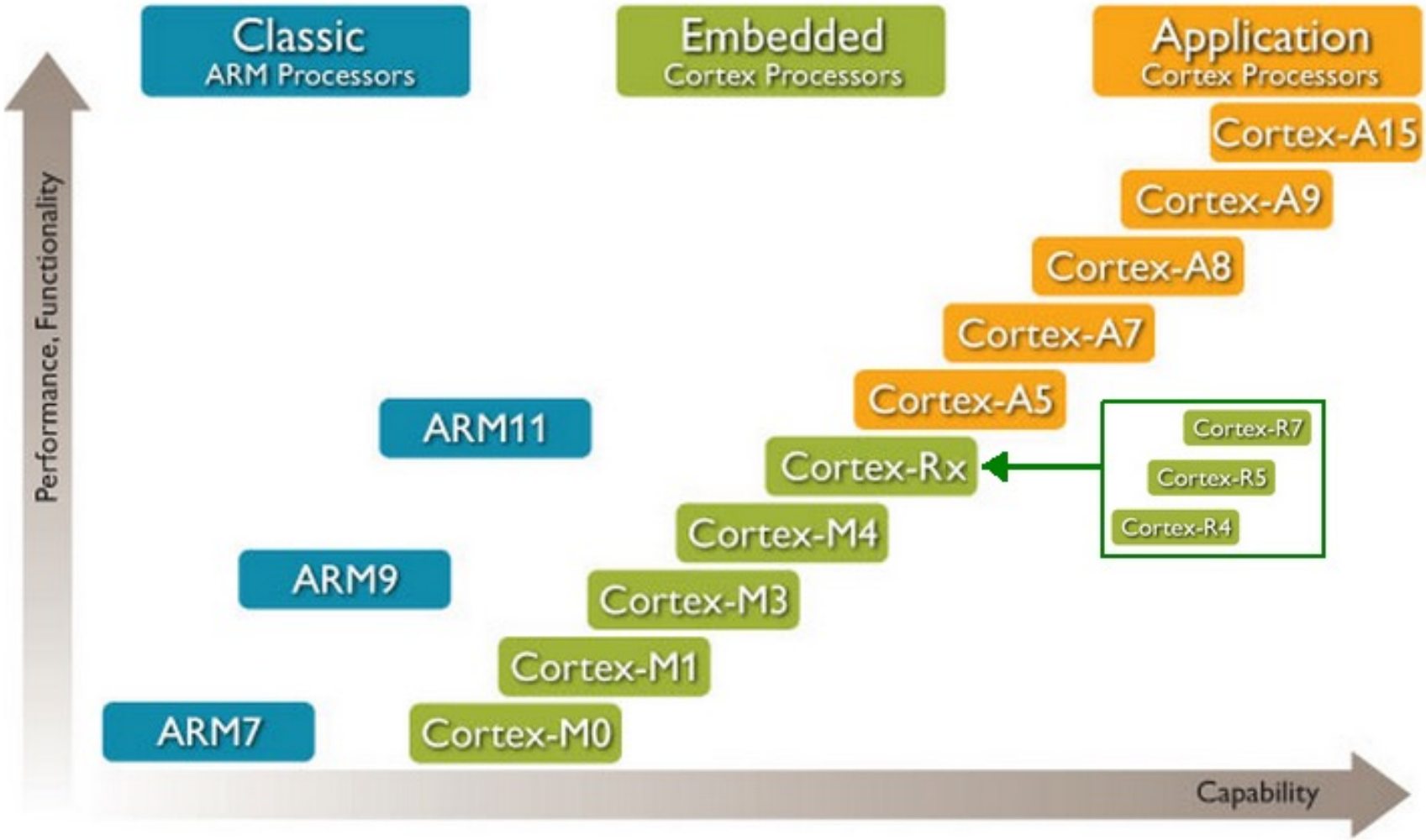
ARM PROCESSOR – APPLICATION DOMAINS

- Home Appliances
- Consumer Electronics
- Home Automation / Security
- Computer components
- Automotive
- Health care
- Military
- Industrial
- Business
- Retail
- Communications
- Avionics
- Banking

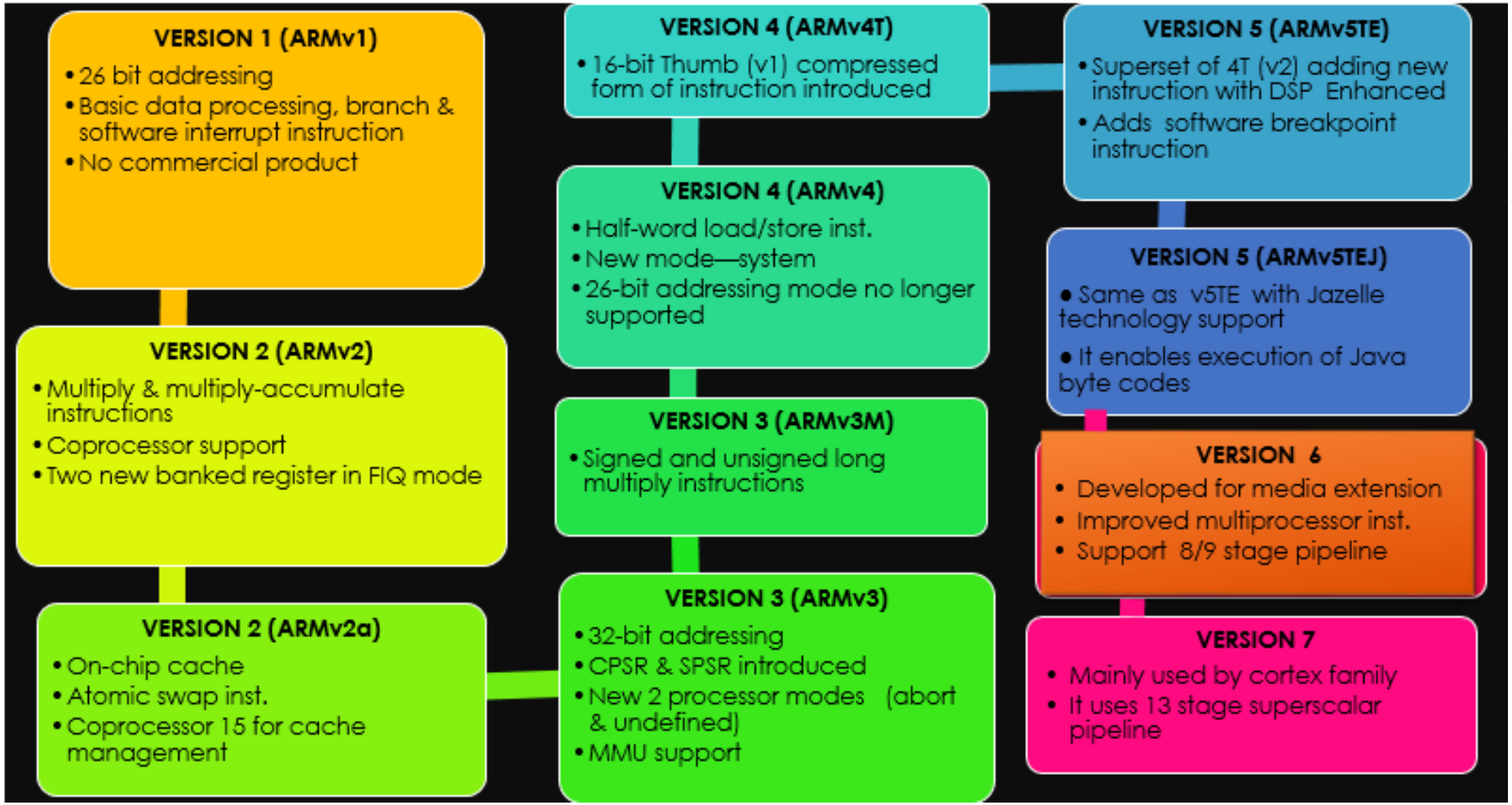
ARM ARCHITECTURE - FEATURES



ARM PROCESSOR - FAMILIES



ARM ARCHITECTURE - VERSIONS



ARM ARCHITECTURE - CURRENT VERSION

Application Profile
ARMv8-A

- 32-bit and 64-bit
- A32, T32 and A64 instruction sets
- Virtual memory system
- Supporting rich operating systems



Real-time Profile
ARMv8-R

- 32-bit
- A32 and T32 instruction sets
- Protected memory system
- Optimized for real-time systems



Microcontroller Profile
ARMv8-M

- 32-bit
- T32 / Thumb® instruction set only
- Protected memory system
- Optimized for microcontroller applications

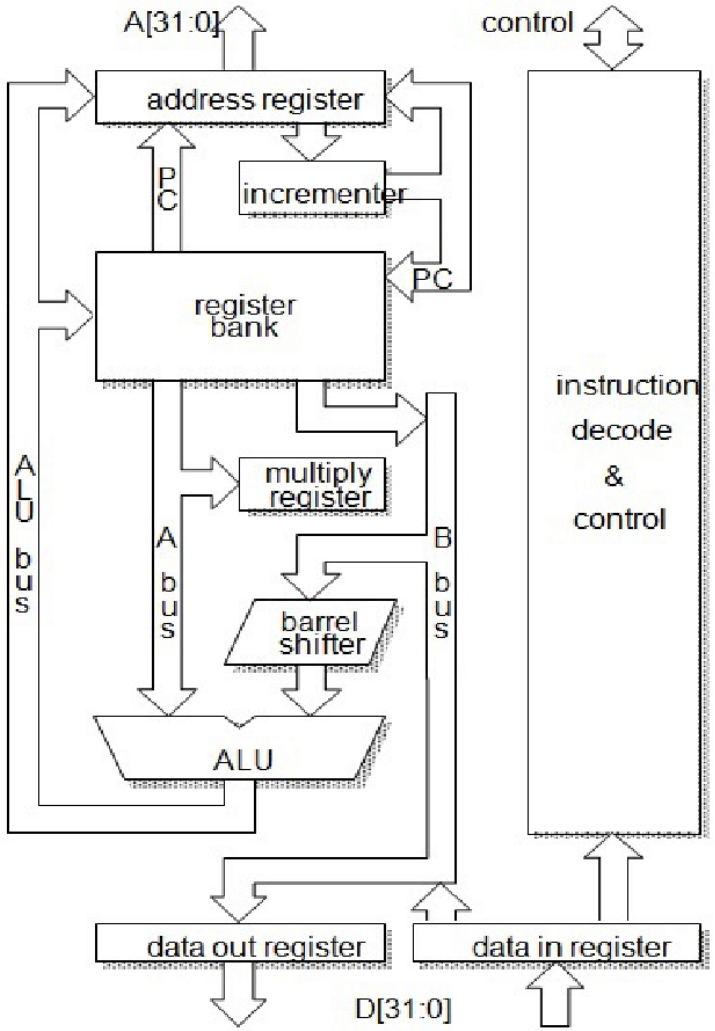


ARM ARCHITECTURE - EVOLUTION

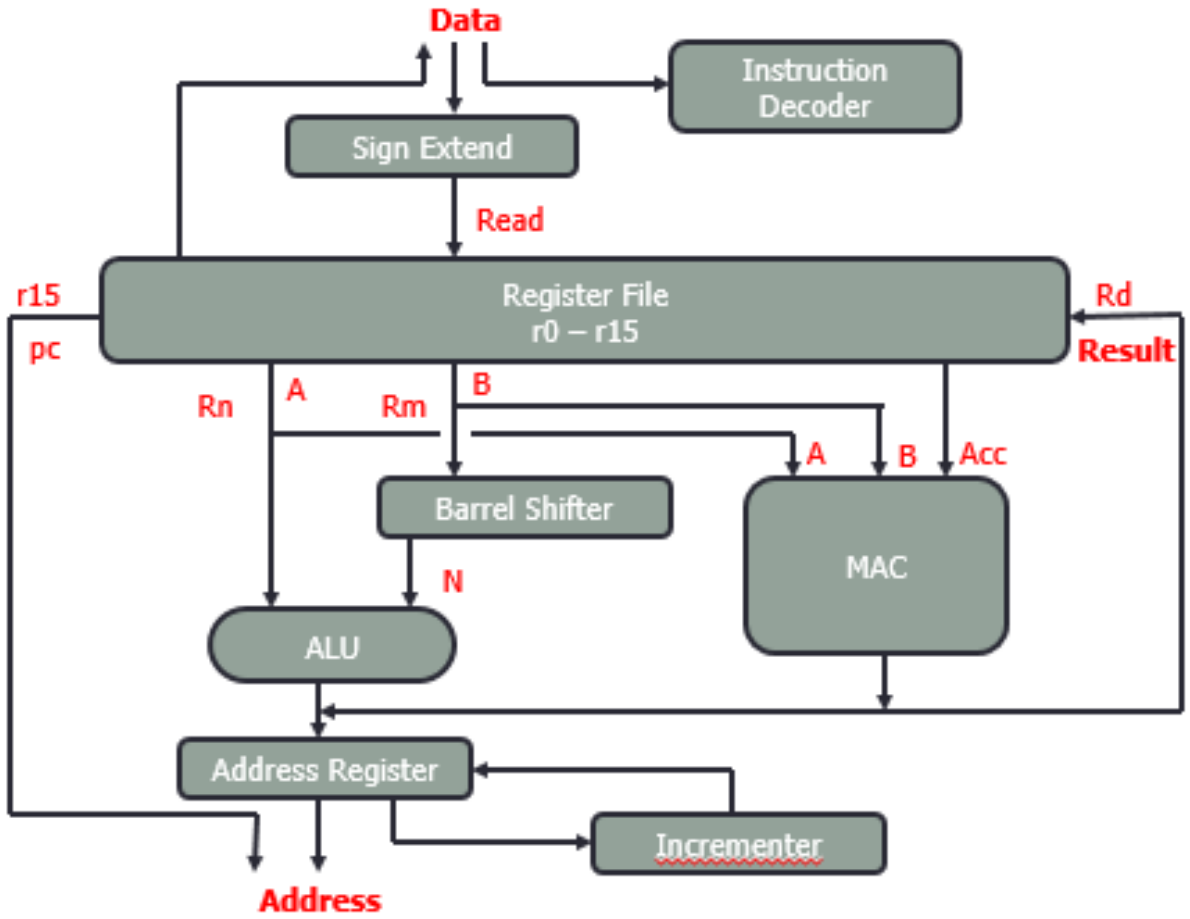
Core	Architecture
ARM1	v1
ARM2	v2
ARM2as, ARM3	v2a
ARM6, ARM600, ARM610	v3
ARM7, ARM700, ARM710	v3
ARM7TDMI, ARM710T, ARM720T, ARM740T	v4T
StrongARM, ARM8, ARM810	v4
ARM9TDMI, ARM920T, ARM940T	V4T
ARM9E-S, ARM10TDMI, ARM1020E	v5TE
ARM10TDMI, ARM1020E	v5TE
ARM11 MPCore, ARM1136J(F)-S, ARM1176JZ(F)-S	v6
Cortex-A/R/M	v7

- ARM{x}{y}{z}{T}{D}{M}{I}{E}{J}{F}{-S}
- x : Family
- y : MMU/Protection unit
- z : Cache
- T : Thumb 16-bit decoder
- D : JTAG Debug
- M : Fast multiplier
- I : Embedded ICE macrocell
- E : DSP Enhanced
- J : Jazelle Technology
- F : Vector floating point unit
- S : Synthesizable version

ARM7 BASIC ARCHITECTURE



ARM CORE DATAFLOW MODEL



ARM CORE DATAFLOW MODEL

- **Instruction Decoder:**
  - The instruction decoder translates instructions before they are executed.
  - The ARM processor, like all RISC processors, uses a load-store architecture.
  - Load instructions copy data from memory to registers in the core, and conversely the store

- **Register File**
  - A storage bank made up of 32-bit registers.
  - Most instructions treat the registers as holding signed or unsigned 32-bit values

ARM CORE DATAFLOW MODEL

- Sign extend hardware
  - Converts signed 8-bit and 16-bit numbers to 32-bit values

- ALU & MAC:
  - Takes the register values Rn and Rm from the A and B buses and computes a result.
  - Data processing instructions write the result in Rd directly to the register file.
  - Load and store instructions use the ALU to generate an address to be held in the address register and broadcast on the Address bus.

ARM CORE DATAFLOW MODEL

- **Source Registers:**
  - ARM instructions have two source registers, Rn &Rm, and a single result or destination register, Rd.
  - Source operands are read from the register file using the internal buses A and B, respectively.

- **Address Register & Incrementer**
  - For load and store instructions the incrementer updates the address register before the core reads or writes the next register value from or to the next sequential memory location.
  - The processor continues executing instructions until an exception or interrupt changes the normal execution flow

ARM CORE DATAFLOW MODEL

- **Barrel Shifter:**
  - register Rm alternatively can be preprocessed in the barrel shifter before it enters the ALU.
  - Provides five types of shift
  - Logic Shift left (LSL)
  - Logic Shift Right (LSR)
  - Arithmetic Shift Right (ASR)
  - Rotate Right(ROR)
  - Rotate Right Extended (RRX)

- After passing through the functional units, the result in Rd is written back to the register file using the Result bus.

ARM PROCESSOR - MODES

- The processor mode determines which registers are active and the access rights to the cpsr register itself.

- Each processor mode is either privileged or nonprivileged:
  - A PRIVILEGED mode allows full read-write access to the CPSR
  - A NONPRIVILEGED mode only allows read access to the control field in the CPDR but still allows read-write access to the condition flags.

- There are seven processor modes in total: six privileged modes (abort, fast interrupt request, interrupt request, supervisor, system, and undefined) and one nonprivileged mode (user).

ARM PROCESSOR - MODES

Exception modes

Mode	Description	
Supervisor (SVC)	Entered on reset and when a Supervisor call instruction (SVC) is executed	Privileged modes
FIQ	Entered when a high priority (fast) interrupt is raised	
IRQ	Entered when a normal priority interrupt is raised	
Abort	Used to handle memory access violations	
Undef	Used to handle undefined instructions	
System	Privileged mode using the same registers as User mode	
User	Mode under which most Applications / OS tasks run	Unprivileged mode

ARM PROCESSOR – REGISTER SET

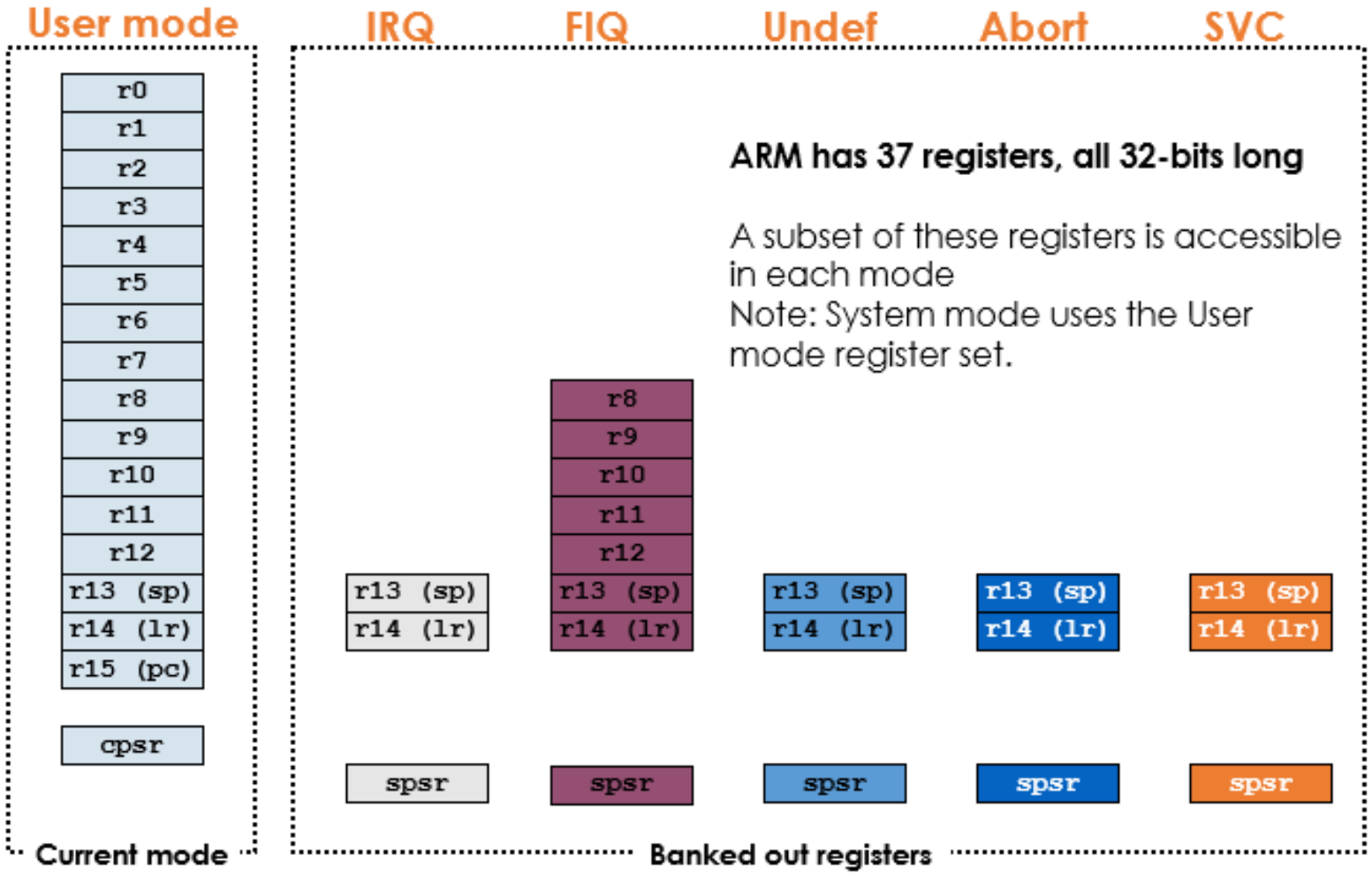
UNBANKED REGISTER

This means that each of them refer to the same 32-bit physical register in all processor modes. They are completely general purpose registers, and can be used whenever an instruction allows GPR to be specified

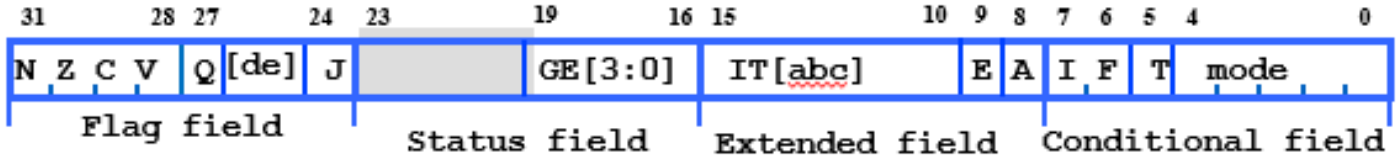
BANKED REGISTER

They are available only when the processor is in a particular mode. Where particular register is intended, without depending on the current processing mode a more specific name is used.

ARM PROCESSOR – REGISTER SET



ARM PROCESSOR – CPSR REGISTER



- Condition code flags
  - N = Negative result from ALU
  - Z = Zero result from ALU
  - C = ALU operation Carried out
  - V = ALU operation oVerflowed
- Sticky Overflow flag - Q flag
  - Indicates if saturation has occurred
- SIMD Condition code bits – GE[3:0]
  - Used by some SIMD instructions
- IF THEN status bits – IT[abcde]
  - Controls conditional execution of Thumb instructions
- T bit
  - T = 0: Processor in ARM state
  - T = 1: Processor in Thumb state
- J bit
  - J = 1: Processor in Jazelle state
- Mode bits
  - Specify the processor mode
- Interrupt Disable bits
  - I = 1: Disables IRQ
  - F = 1: Disables FIQ
- E bit
  - E = 0: Data load/store is little endian
  - E = 1: Data load/store is bigendian
- A bit
  - A = 1: Disable imprecise data aborts

Mode bits	
M[4:0]	Mode
0b10000	User
0b11111	System
0b10001	FIQ
0b10010	IRQ
0b10011	Supervisor
0b10111	Abort
0b11011	Undefined

PIC

FEATURES

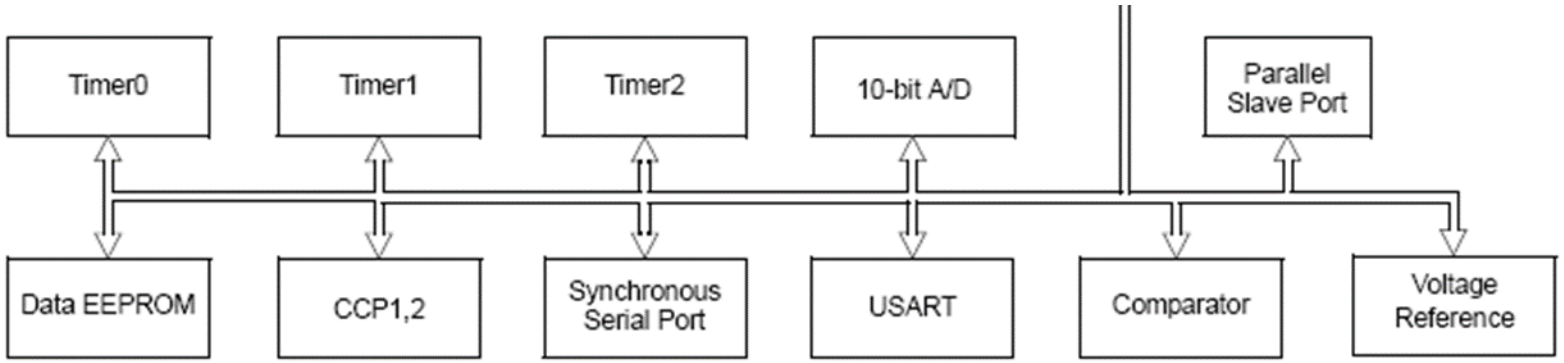
- This powerful (200 nanosecond instruction execution) yet easy-to-program (RISC instructions).
- CMOS FLASH-based 8-bit microcontroller packs Microchip's powerful PIC® architecture into an 40-pin dual in-line (DIP)
- The PIC16F877A features 256 bytes of EEPROM data memory, self programming, an ICD, 2 Comparators, 8 channels of 10-bit Analog-to-Digital (A/D) converter, 2 input capture/output compare/PWM functions, the synchronous serial port can be configured as either 3-wire Serial Peripheral Interface (SPI™) or the 2-wire Inter-Integrated Circuit (I²C™) bus and a Universal Asynchronous Receiver Transmitter (USART).

PIC

MEMORY

- 8 K of flash program ROM (14-bits wide).
- Each of the 35 RISC instructions fits in just one 14-bit wide location, permitting fast (200 ns) instruction execution time, if microcontroller is clocked at its maximum allowable clock rate of 20 MHz.
- 368 bytes of volatile data RAM “File Register” locations (8-bits wide) for temporary storage of data.
- 256 bytes of nonvolatile EEROM for storage of calibration data, user-customized option settings, etc.

PIC16F877A HARDWARE BLOCK DIAGRAM



Device	Program Flash	Data Memory	Data EEPROM
PIC16F874A	4K words	192 Bytes	128 Bytes
PIC16F877A	8K words	368 Bytes	256 Bytes

PIC

Special Functions of PIC16F877A I/O Pins

- Three hardware timers: Timer 0 (RA4/TOCK1 pin), Timer 1 (RC0/T1OSO/TICKI and RC1/T1OSI pins), and Timer 2 (no external pins are associated with this timer – this timer can be set to periodically interrupt via the PR2 period register).

- 10-bit, 8-channel analog-to-digital converter (analog inputs on Ports A and E)

- 8-bit parallel slave port (PSP) for making the PIC look like an addressable 8-bit peripheral register on a host computer's bus (Port D for the 8-bit data bus connection and PORT E for the RD\, WR\, and CS\ bus control lines).

PIC

Special Functions of PIC16F877A I/O Pins

- Input Capture – Output Compare – Pulse Width Modulation pins (RC2/CCP1 and RC1/T1OSI/CCP2).
- Synchronous serial data port (SDI, SDO, and SCK pins) used for implementing master and slave serial peripheral interface SPI and I2C serial interfaces.
- Asynchronous serial data port, or Universal Asynchronous Receiver-Transmitter, (RC6/TX and RC7/RX pins).
- Two analog comparators with programmable analog reference voltage (PORT A pins are used for the analog comparator inputs and also the digital comparator outputs.

PIC

	8051	PIC	AVR	ARM
Bus width	8-bit for standard core	8/16/32-bit	8/32-bit	32-bit mostly also available in 64-bit
Communication Protocols	UART, USART,SPI,I2C	PIC, UART, USART, LIN, CAN, Ethernet, SPI, I2S	UART, USART, SPI, I2C, (special purpose AVR support CAN, USB, Ethernet)	UART, USART, LIN, I2C, SPI, CAN, USB, Ethernet, I2S, DSP, SAI (serial audio interface), IrDA
Speed	12 Clock/instruction cycle	4 Clock/instruction cycle	1 clock/ instruction cycle	1 clock/ instruction cycle
Memory	ROM, SRAM, FLASH	SRAM, FLASH	Flash, SRAM, EEPROM	Flash, SDRAM, EEPROM
ISA	CISC	Some feature of RISC	RISC	RISC
Memory Architecture	Von Neumann architecture	Harvard architecture	Modified	Modified Harvard architecture
Power Consumption	Average	Low	Low	Low

PIC

	8051	PIC	AVR	ARM
Families	8051 variants	PIC16,PIC17, PIC18, PIC24, PIC32	Tiny, Atmega, Xmega, special purpose AVR	ARMv4,5,6,7 and series
Community	Vast	Very Good	Very Good	Vast
Manufacturer	NXP, Atmel, Silicon Labs, Dallas, Cyprus, Infineon, etc.	Microchip Average	Atmel	Apple, Nvidia, Qualcomm, Samsung Electronics, and TI etc.
Cost (as compared to features provide)	Very Low	Average	Average	Low
Other Feature	Known for its Standard	Cheap	Cheap, effective	High speed operation
Popular Microcontrollers	AT89C51, P89v51, etc.	PIC18fXX8, PIC16f88X, PIC32MXX	Atmega8, 16, 32, Arduino Community	LPC2148, ARM Cortex-M0 to ARM Cortex-M7, etc.

THANK YOU

